Details of Data Analyst

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Group: 2

Road Accident Analysis in India

Exploratory Data Analysis, Visualization & Statistical Inference Report

1. Introduction & Problem Context

Road accidents remain one of the leading causes of fatalities and injuries in India. Understanding the factors contributing to accidents—such as speed, road type, lighting conditions, alcohol involvement, and vehicle density—is crucial for designing effective road safety policies. This project aims to perform a comprehensive exploratory data analysis and statistical investigation on a large-scale Indian road accident dataset to uncover meaningful patterns, trends, and relationships within the data.

The analysis follows a structured data science approach beginning from data loading and cleaning, followed by visualization-based exploration and finally inferential statistical testing to support data-driven conclusions.

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.preprocessing import StandardScaler
```

---

2. Dataset Overview and Structure

The dataset contains **22,500** accident records with **18** attributes, representing accident identifiers, temporal information, geographical details, road and environmental conditions, speed-related metrics, injury severity, and economic damage.

Key observations from the dataset structure:

- Both categorical (state, city, road type, weather, light condition) and numerical variables (speed, casualties, injuries, damage cost) are present.
- This mixed data type structure allows for univariate, bivariate, and multivariate analysis. The dataset size is sufficiently large to apply statistical inference techniques reliably.

Functions used:

- **pd.read_csv()** to load structured data
- **df.info()** and **df.describe()** to understand schema, data types, and statistical summaries

These functions are essential for initial data familiarity and validation.

In [3]:
```
df = pd.read_csv("indian_road_accidents.csv")
df.head()
```

Out[3]:

	accident_id	date	time	state	city	road_type	weather	light_con
0	ACC2020051000000	2020-05-10	10:42	West Bengal	Kolkata	Urban Road	Clear	Da
1	ACC2023081000001	2023-08-10	10:46	Gujarat	Vadodara	Urban Road	Clear	Dawn
2	ACC2023051000002	2023-05-10	17:39	Karnataka	Mysuru	Urban Road	Clear	Da
3	ACC2025011200003	2025-01-12	21:00	Karnataka	Bengaluru	National Highway	Clear	Night
4	ACC2020081800004	2020-08-18	10:36	Chhattisgarh	Raipur	Expressway	Fog	Da

In [4]:
```
df.shape
```

Out[4]:
```
(22500, 18)
```

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22500 entries, 0 to 22499
Data columns (total 18 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   accident_id         22500 non-null  object
 1   date                22500 non-null  object
 2   time                22500 non-null  object
 3   state               22500 non-null  object
 4   city                22500 non-null  object
 5   road_type           22500 non-null  object
 6   weather             20700 non-null  object
 7   light_condition     21375 non-null  object
 8   speed_limit_kmph    22500 non-null  int64
 9   avg_speed_est_kmph  21150 non-null  float64
 10  vehicles_involved   22500 non-null  int64
 11  casualties_total    22500 non-null  int64
 12  fatalities          22500 non-null  int64
 13  serious_injuries    22500 non-null  int64
 14  minor_injuries      22500 non-null  int64
 15  alcohol_involved    22050 non-null  float64
 16  hit_and_run         22050 non-null  float64
 17  damage_cost_inr     21600 non-null  float64
dtypes: float64(4), int64(6), object(8)
memory usage: 3.1+ MB
```

In [6]:
```
df.describe()
```

Out[6]:

	speed_limit_kmph	avg_speed_est_kmph	vehicles_involved	casualties_total	fatali
count	22500.000000	21150.000000	22500.000000	22500.000000	22500.000
mean	67.846667	61.518137	1.731200	2.780044	0.138
std	19.395013	23.632254	1.231364	2.577795	0.611
min	40.000000	14.400000	1.000000	0.000000	0.000
25%	50.000000	44.500000	1.000000	1.000000	0.000
50%	60.000000	57.400000	1.000000	2.000000	0.000
75%	80.000000	75.500000	2.000000	4.000000	0.000
max	120.000000	295.401231	12.000000	42.000000	17.000

In [7]:
```
df.dtypes
```

Out[7]:
```
accident_id            object
date                   object
time                   object
state                  object
city                   object
road_type              object
weather                object
light_condition        object
speed_limit_kmph        int64
avg_speed_est_kmph    float64
vehicles_involved       int64
casualties_total        int64
fatalities              int64
serious_injuries        int64
minor_injuries          int64
alcohol_involved      float64
hit_and_run           float64
damage_cost_inr       float64
dtype: object
```

---

3. Data Quality Assessment

Duplicate Records

The dataset was checked for duplicate rows using the **duplicated()** function. No duplicate records were found, confirming data uniqueness at the record level.

Why this matters:

Duplicate entries can bias statistical results, especially frequency-based analyses such as count plots and probability distributions.

In [8]:
```
df.duplicated().sum()
```

Out[8]:
```
np.int64(0)
```

In [9]:
```
df.drop_duplicates(inplace=True)
```

Missing Value Analysis

Several columns such as weather, light_condition, avg_speed_est_kmph, alcohol_involved, and damage_cost_inr contained missing values.

Approach used:

- **Numerical columns** → Mean imputation using **fillna(df[col].mean())**

- **Categorical columns** → Mode imputation using **fillna(df[col].mode()[0])**

Why this approach:

- **Mean** imputation preserves the overall distribution for continuous variables.
- **Mode** imputation retains the most frequently occurring category, suitable for nominal data.

This approach avoids excessive row deletion, which could reduce dataset representativeness. After imputation, the dataset contained zero missing values, making it suitable for downstream analysis.

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
accident_id              0
date                     0
time                     0
state                    0
city                     0
road_type                0
weather               1800
light_condition       1125
speed_limit_kmph         0
avg_speed_est_kmph    1350
vehicles_involved        0
casualties_total         0
fatalities               0
serious_injuries         0
minor_injuries           0
alcohol_involved       450
hit_and_run            450
damage_cost_inr        900
dtype: int64
```

In [11]:
```
num_cols = df.select_dtypes(include=np.number).columns
cat_cols = df.select_dtypes(include="object").columns
```

In [46]:
```
for col in num_cols:
    df[col] = df[col].fillna(df[col].mean())

for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
```

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
accident_id           0
date                  0
time                  0
state                 0
city                  0
road_type             0
weather               0
light_condition       0
speed_limit_kmph      0
avg_speed_est_kmph    0
vehicles_involved     0
casualties_total      0
fatalities            0
serious_injuries      0
minor_injuries        0
alcohol_involved      0
hit_and_run           0
damage_cost_inr       0
dtype: int64
```

---

4. Outlier Detection and Treatment

Detection

Boxplots were generated for all numerical variables using **sns.boxplot().**

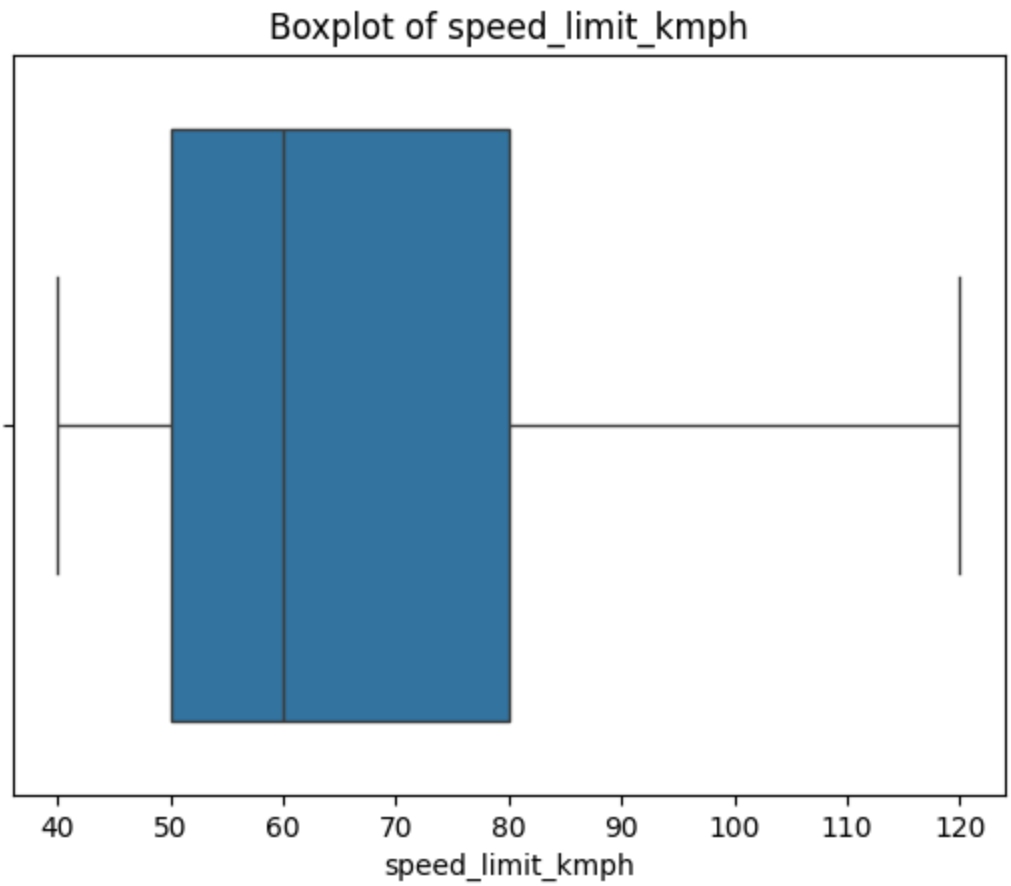
Why boxplots:

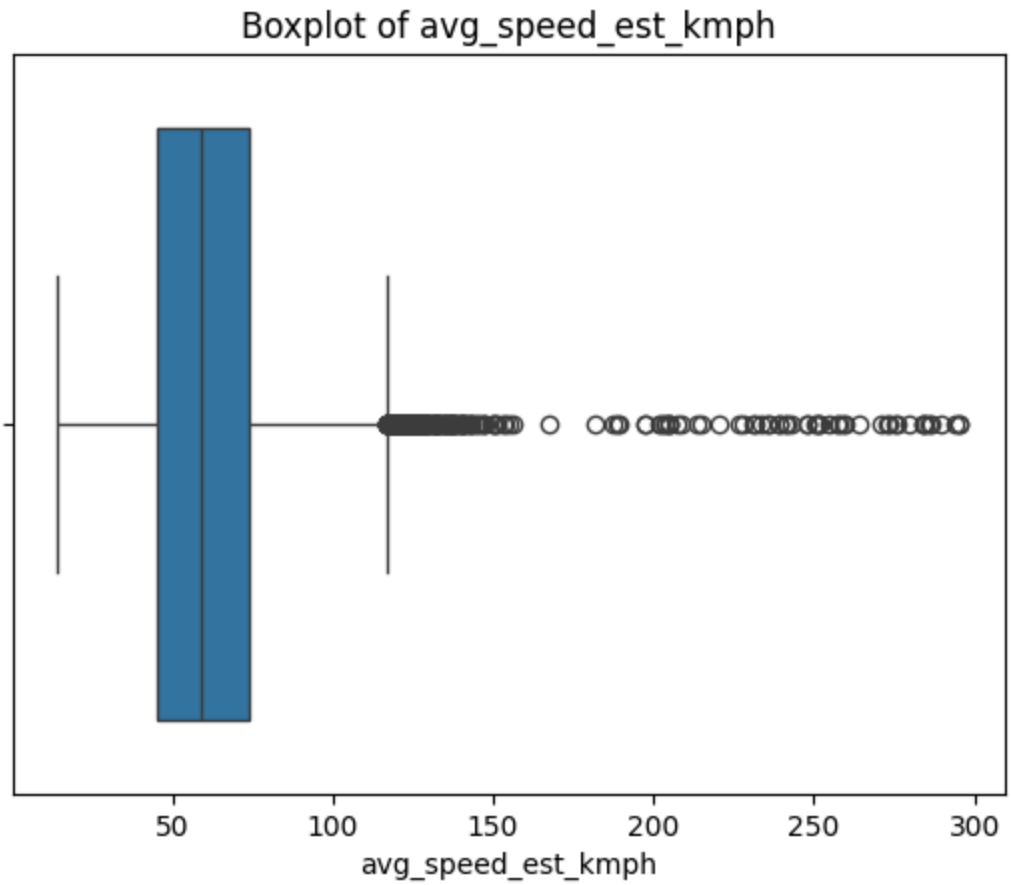
- They visually highlight extreme values using the **interquartile range (IQR).**
- They are especially effective for skewed distributions such as accident costs or injury counts.

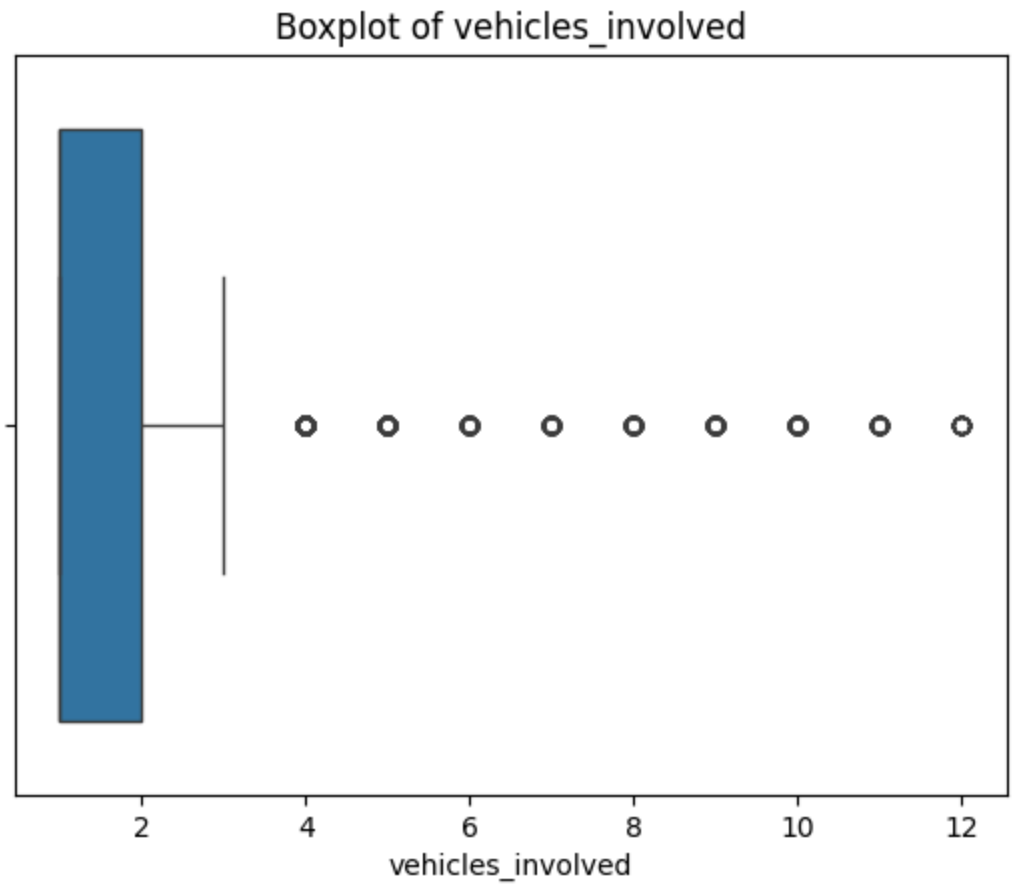
Observations:

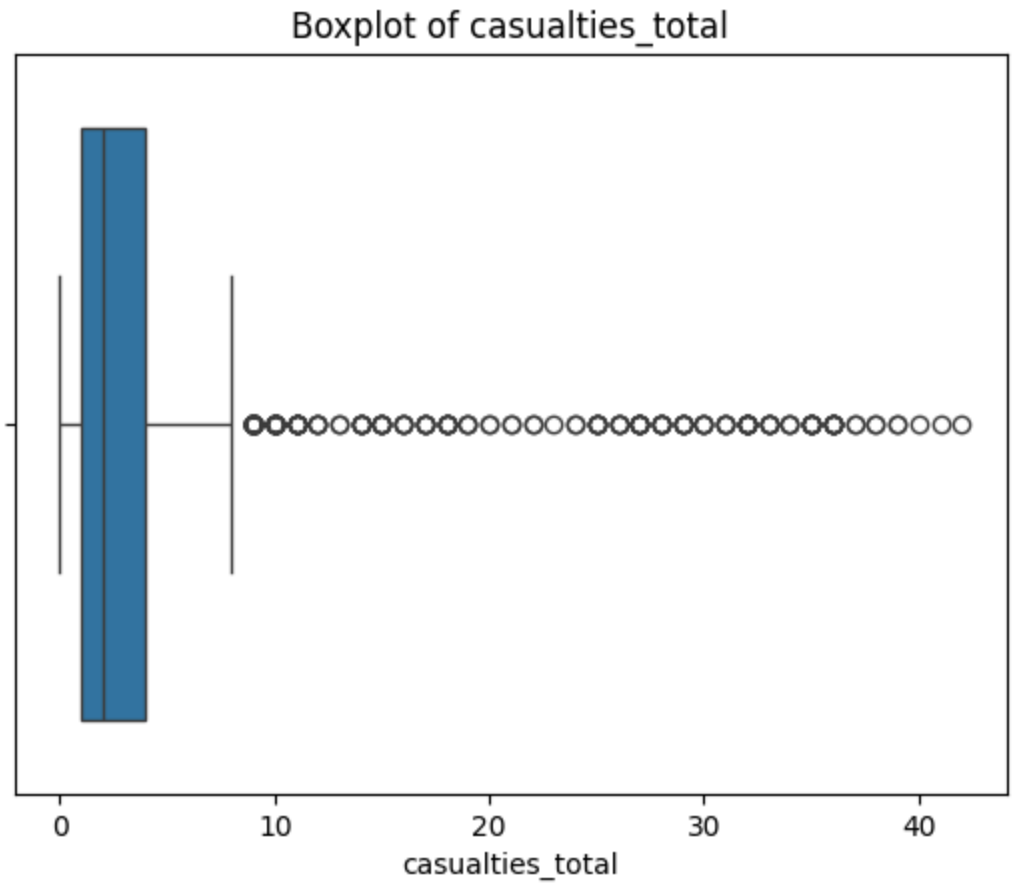
- Variables like avg_speed_est_kmph, casualties_total, damage_cost_inr showed significant outliers.
- These outliers represent rare but extreme accident events.

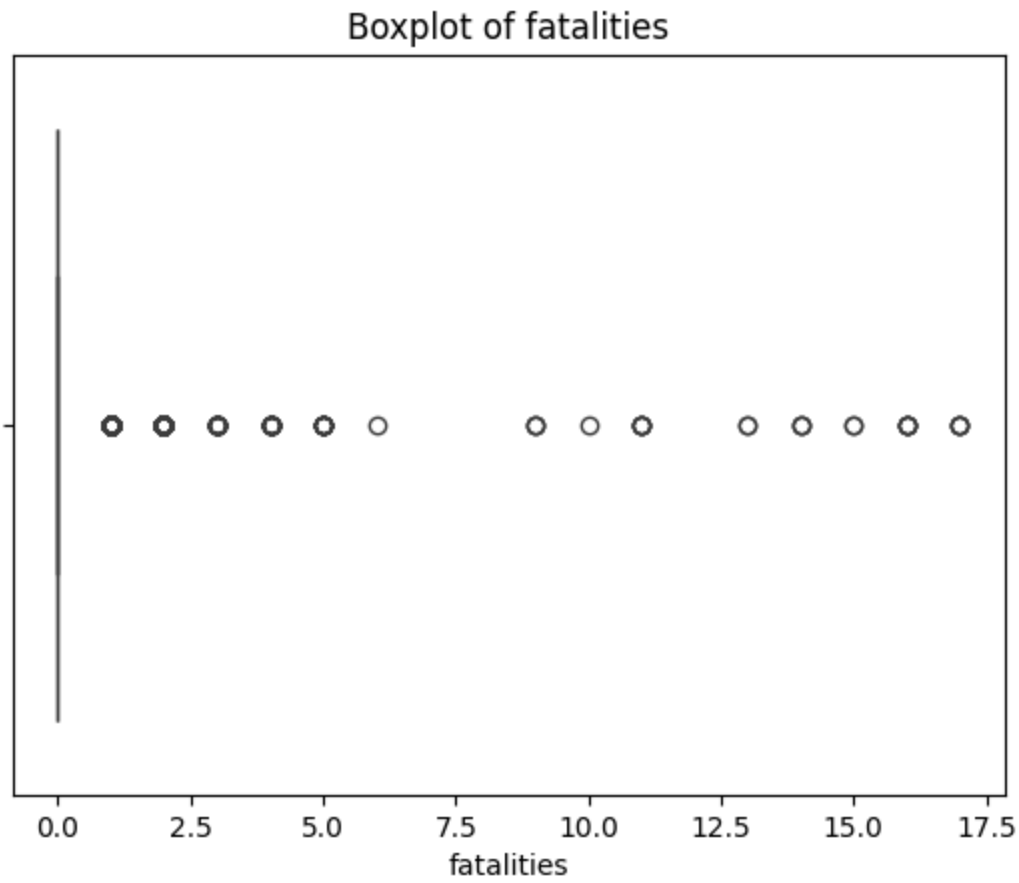
In [14]:
```
for col in num_cols:
    sns.boxplot(x=df[col])
    plt.title(f"Boxplot of {col}")
    plt.show()
```

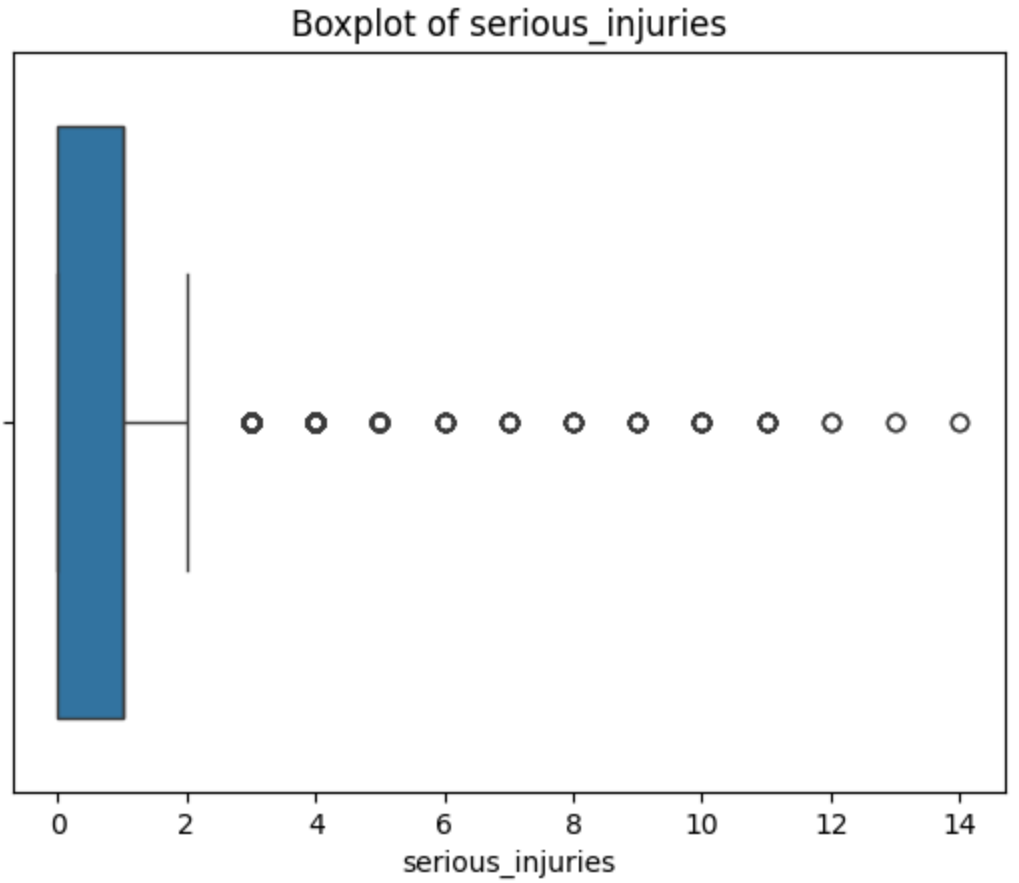

Boxplot of speed_limit_kmph
40
50
60
70
80
90
100
110
120
speed_limit_kmph

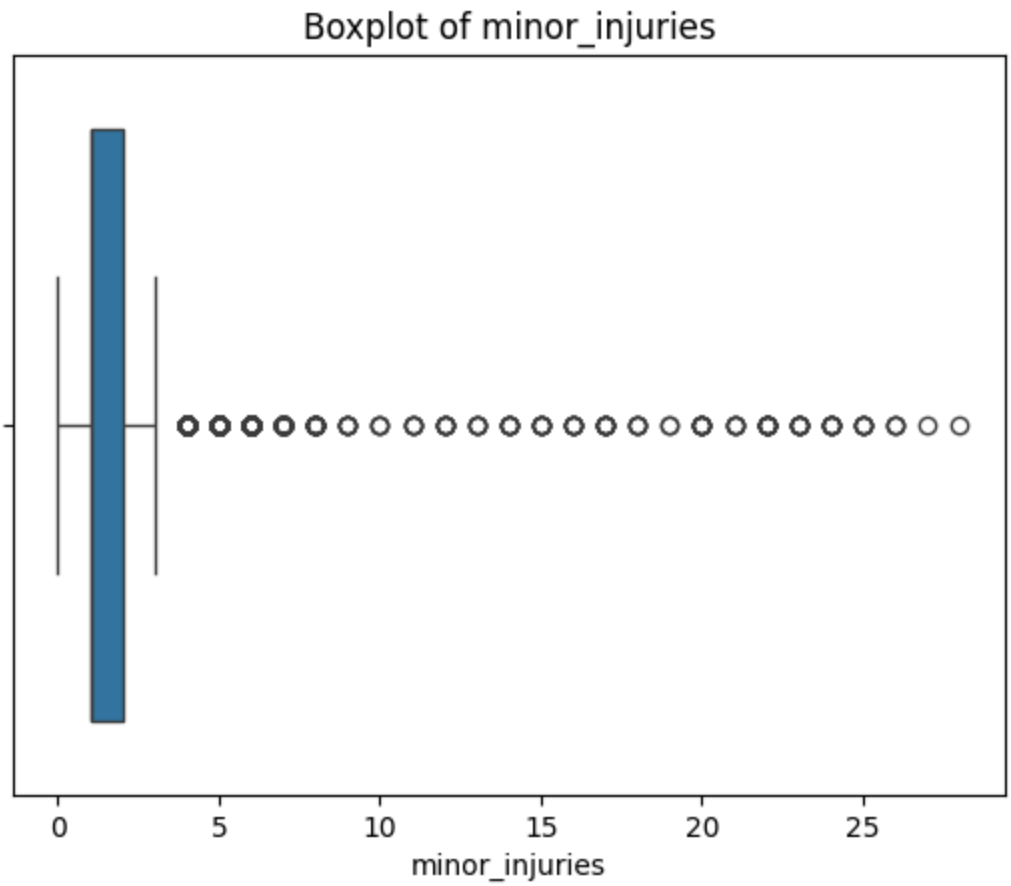

Boxplot of avg_speed_est_kmph
50
100
150
200
250
300
avg_speed_est_kmph

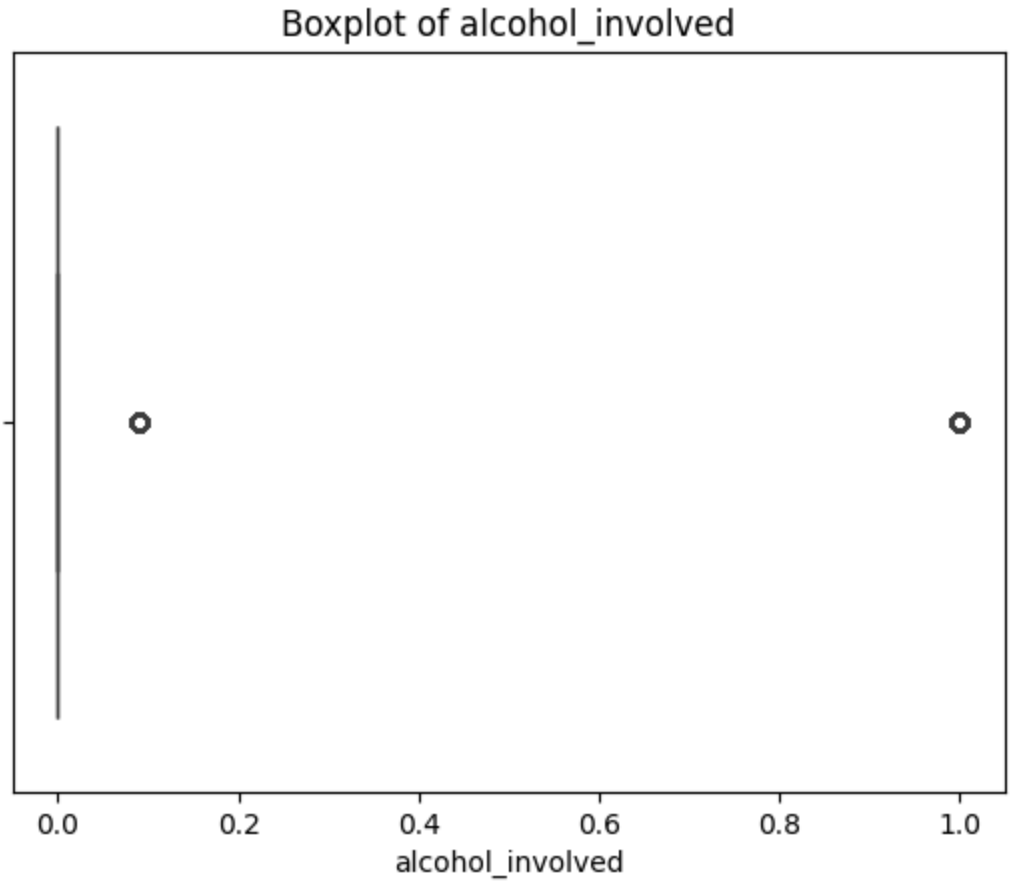

Boxplot of vehicles_involved
2
4
6
8
10
12
vehicles_involved

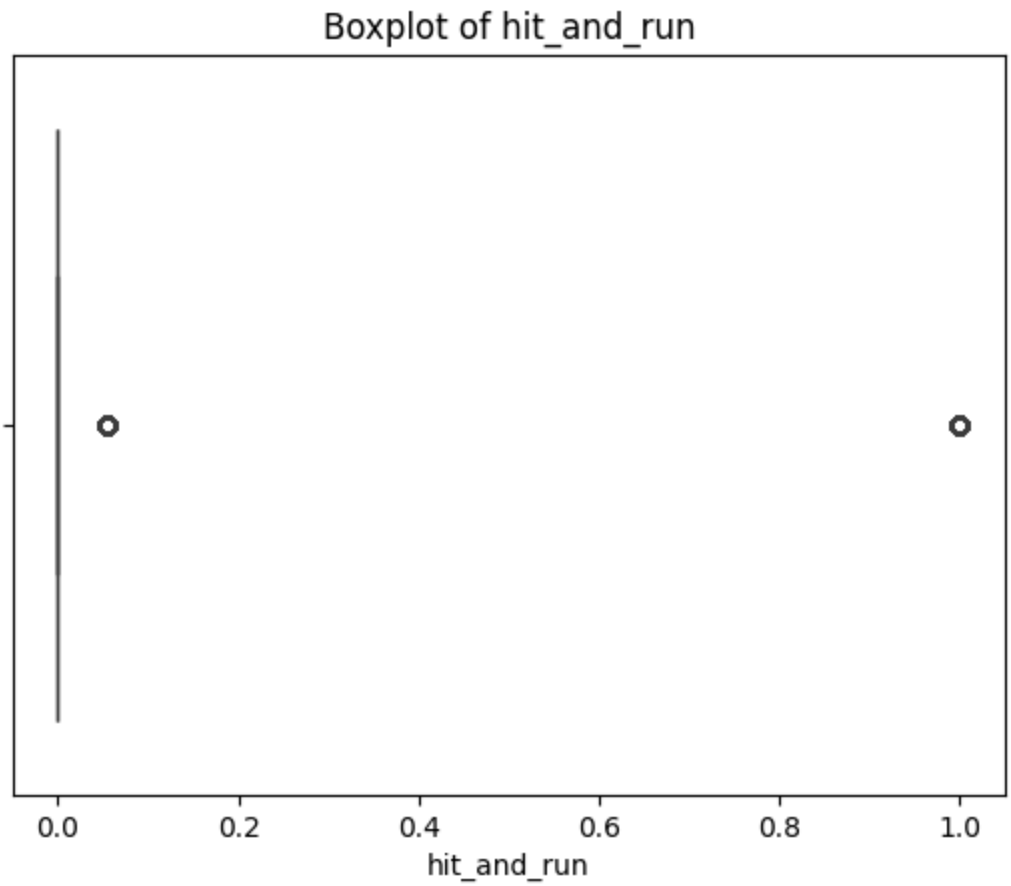

Boxplot of casualties_total
0
10
20
30
40
casualties_total

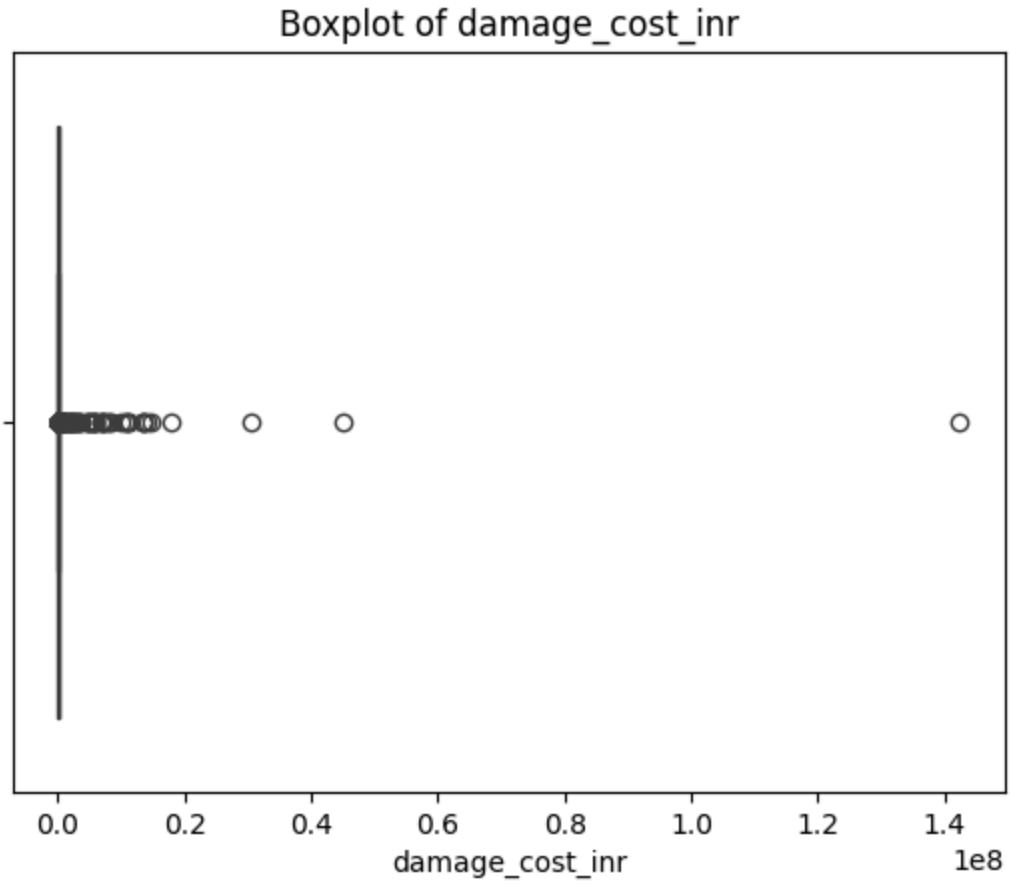

Boxplot of fatalities
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
fatalities


Boxplot of serious_injuries
0
2
4
6
8
10
12
14
serious_injuries


Boxplot of minor_injuries
0
5
10
15
20
25
minor_injuries


Boxplot of alcohol_involved
0.0
0.2
0.4
0.6
0.8
1.0
alcohol_involved


Boxplot of hit_and_run
0.0
0.2
0.4
0.6
0.8
1.0
hit_and_run


Boxplot of damage_cost_inr
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1e8
damage_cost_inr

Treatment

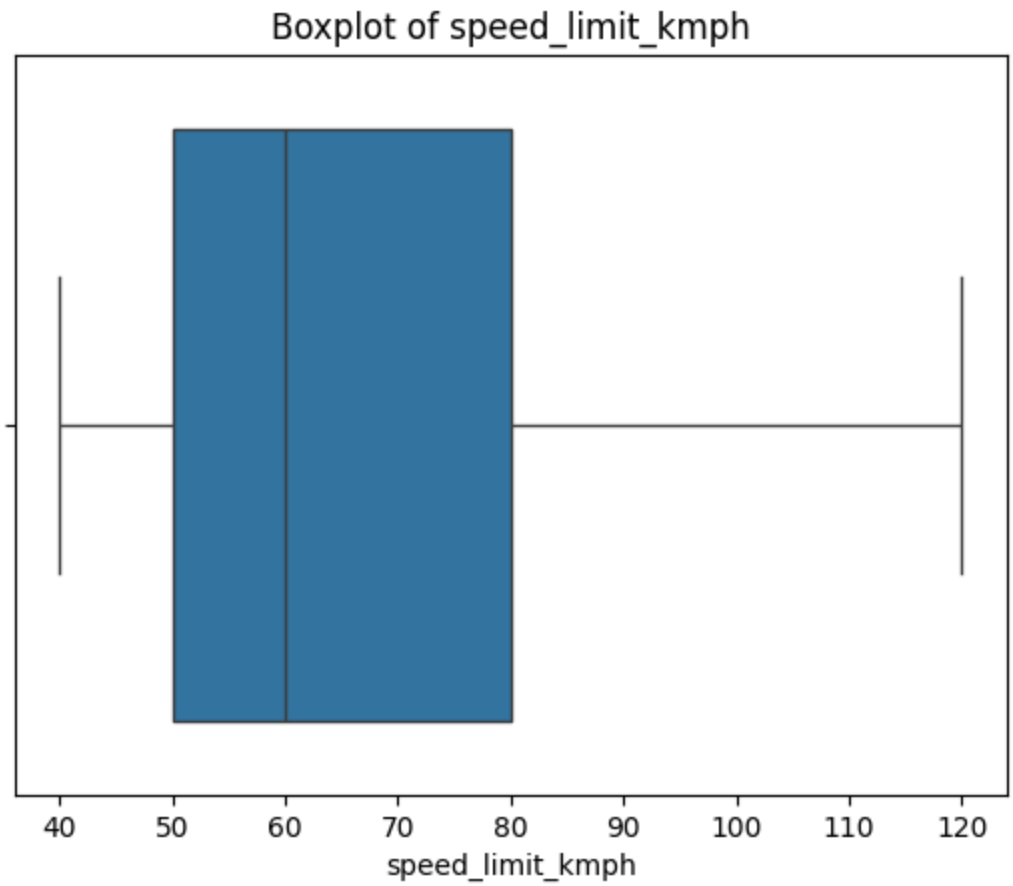
The IQR capping method was applied:
Values below **Q1 − 1.5×IQR** and above **Q3 + 1.5×IQR** were capped.

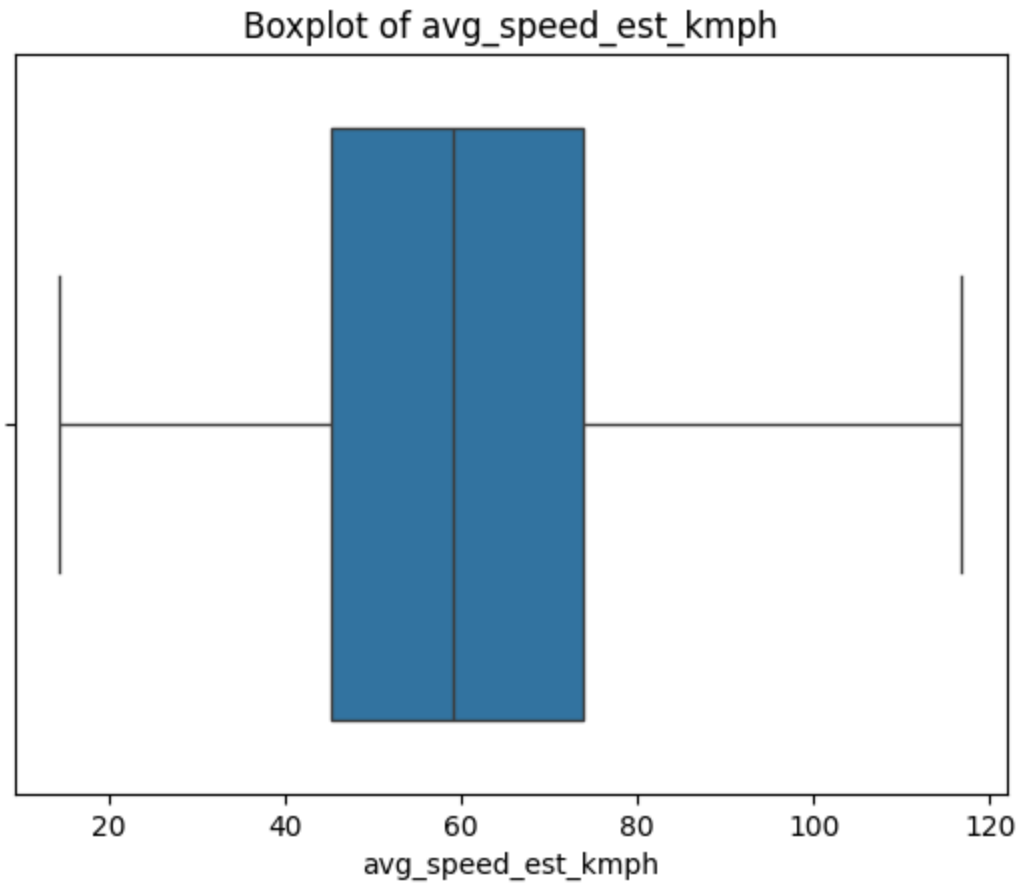
Why capping instead of removal:

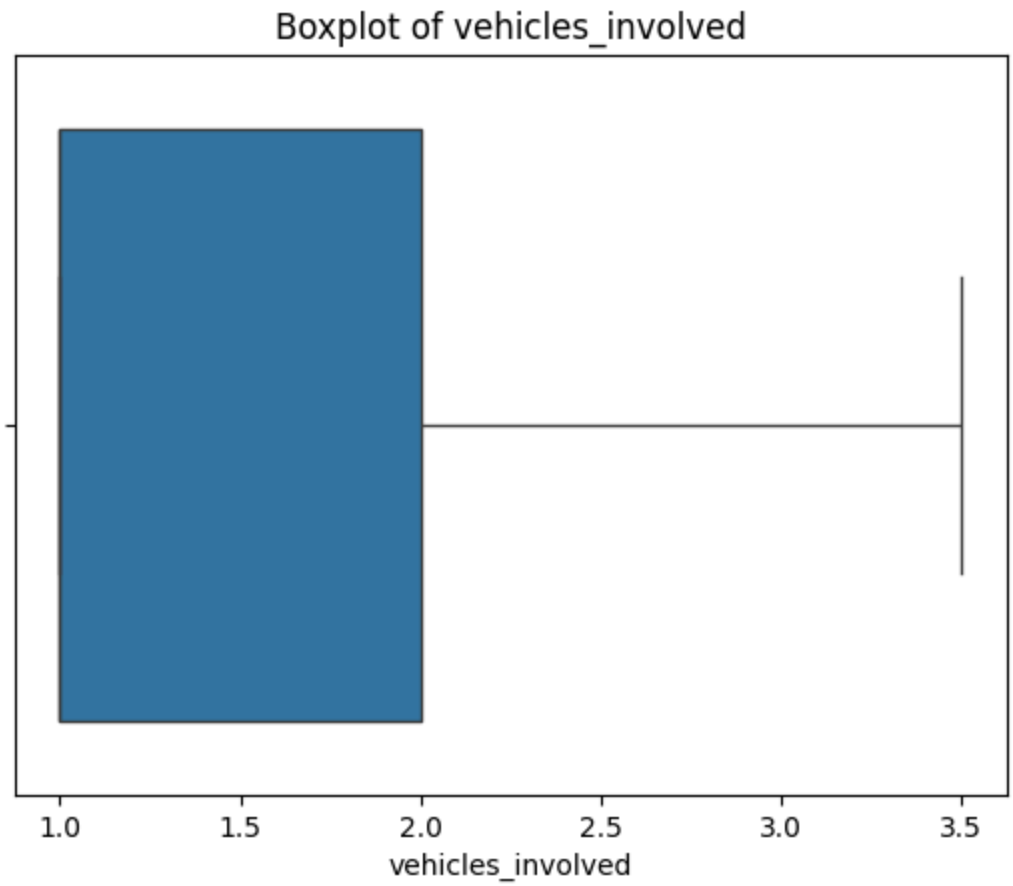
- Removing extreme accidents would discard critical real-world events.
- Capping limits their influence without eliminating them, preserving analytical realism.

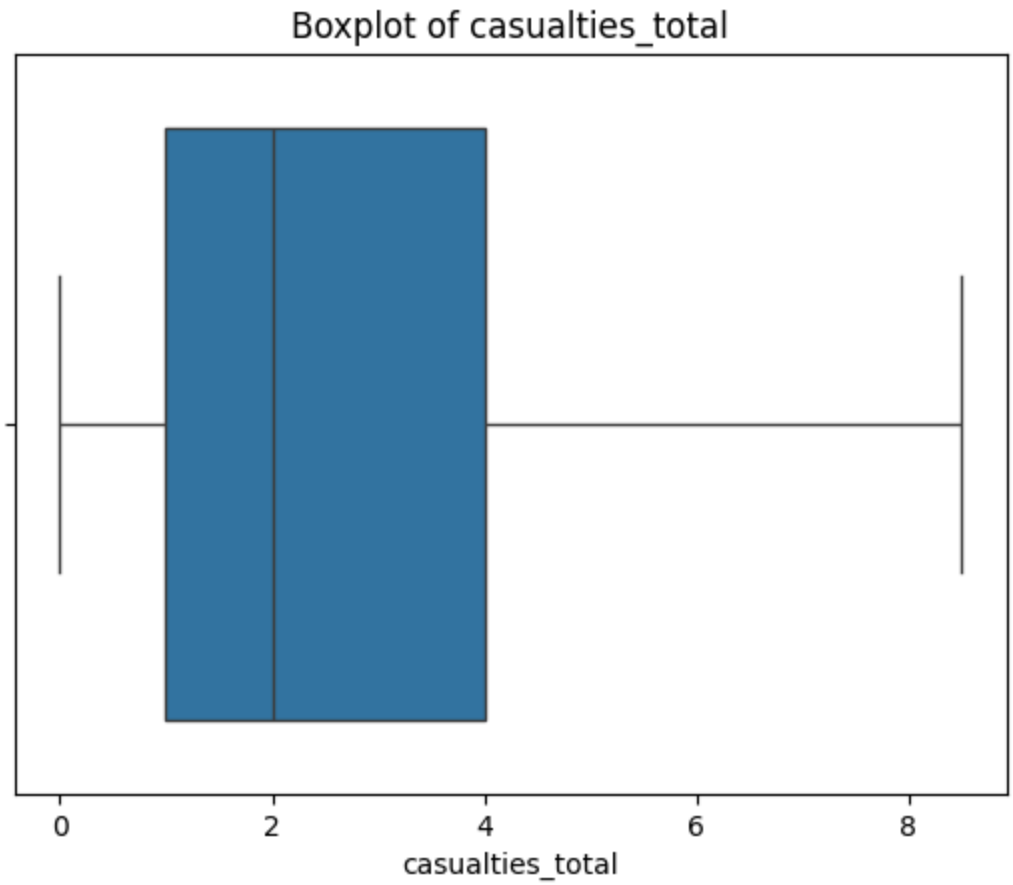
In [15]:
```
for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    df[col] = np.where(df[col] < lower, lower, df[col])
    df[col] = np.where(df[col] > upper, upper, df[col])
```

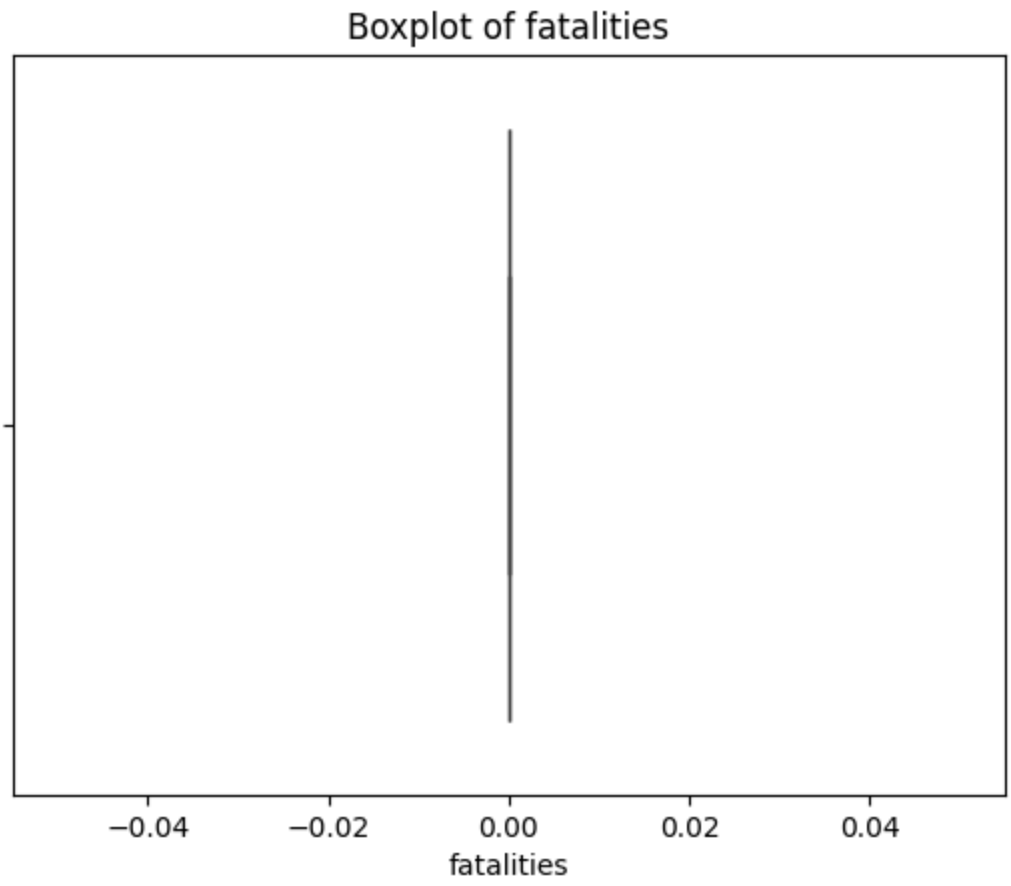
In [16]:
```
for col in num_cols:
    sns.boxplot(x=df[col])
    plt.title(f"Boxplot of {col}")
    plt.show()
```

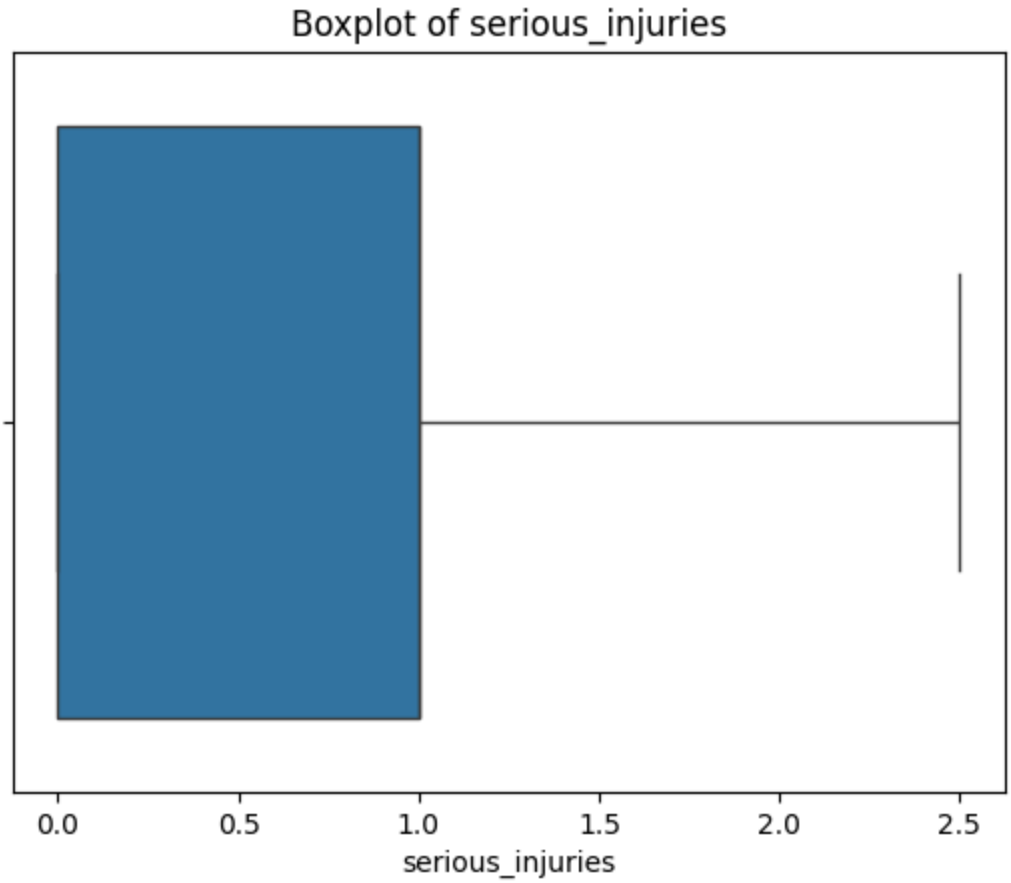

Boxplot of speed_limit_kmph
40
50
60
70
80
90
100
110
120
speed_limit_kmph

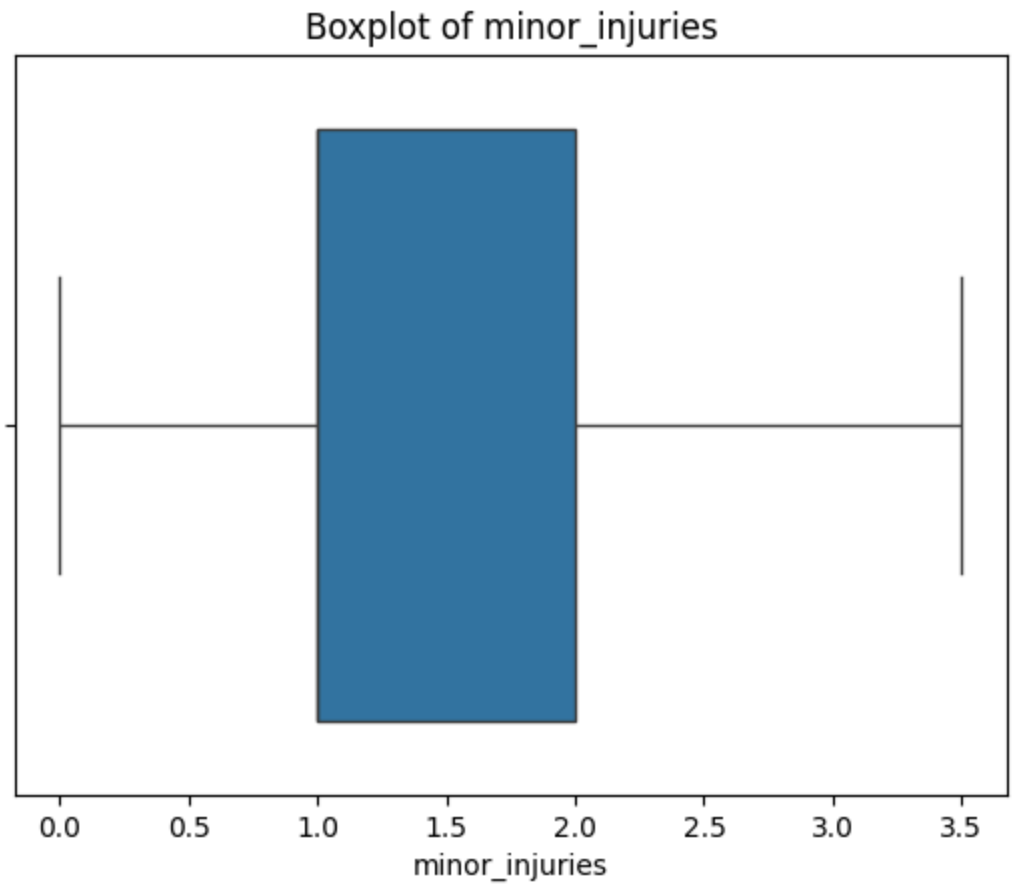

Boxplot of avg_speed_est_kmph
20
40
60
80
100
120
avg_speed_est_kmph

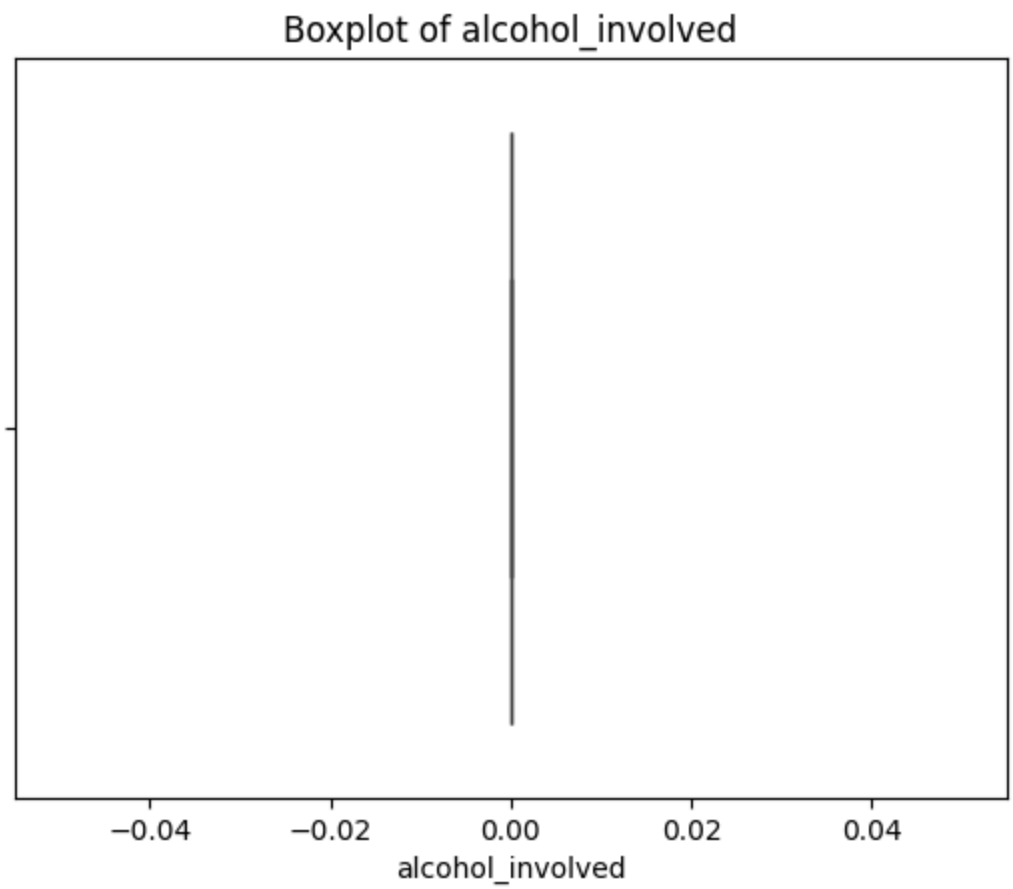

Boxplot of vehicles_involved
1.0
1.5
2.0
2.5
3.0
3.5
vehicles_involved

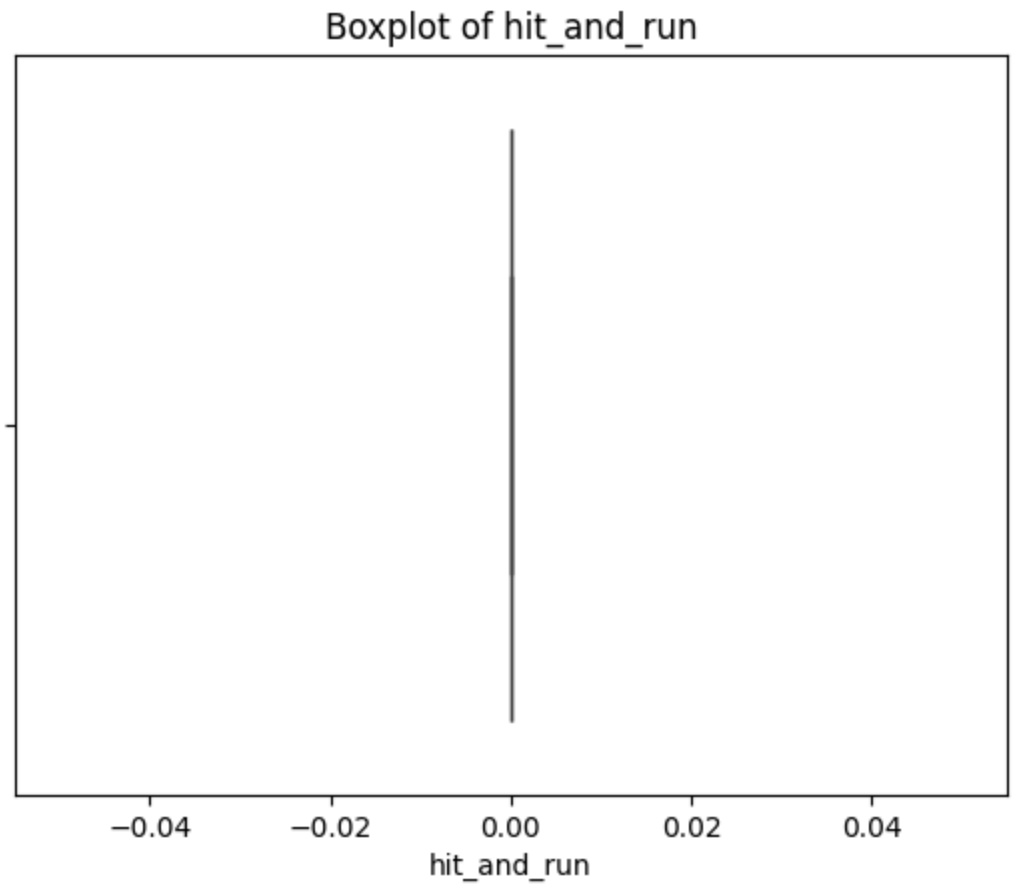

Boxplot of casualties_total
0
2
4
6
8
casualties_total

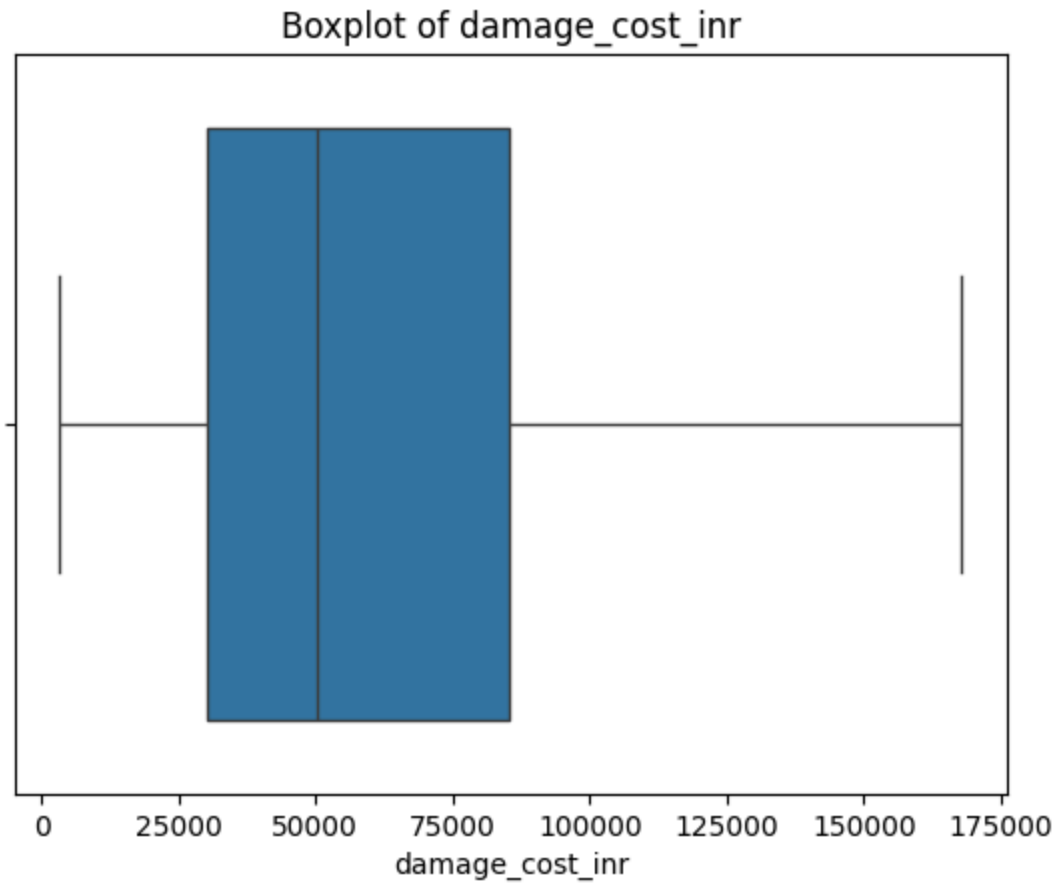

Boxplot of fatalities
−0.04
−0.02
0.00
0.02
0.04
fatalities


Boxplot of serious_injuries
0.0
0.5
1.0
1.5
2.0
2.5
serious_injuries


Boxplot of minor_injuries
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
minor_injuries


Boxplot of alcohol_involved
−0.04
−0.02
0.00
0.02
0.04
alcohol_involved


Boxplot of hit_and_run
−0.04
−0.02
0.00
0.02
0.04
hit_and_run


Boxplot of damage_cost_inr
0
25000
50000
75000
100000
125000
150000
175000
damage_cost_inr

---

5. Data Standardization

All numerical features were standardized using **StandardScaler.**

Why standardization was necessary:

- Variables are measured on different scales (speed in km/h, cost in INR, injury counts).
- Standardization ensures fair comparison and improves visualization symmetry.
- It is essential before correlation analysis and statistical modeling.

In [17]:
```
scaler = StandardScaler()
df[num_cols] = scaler.fit_transform(df[num_cols])
```

---

6. Univariate Analysis and Insights

Histograms
Histograms were plotted for each numerical variable using **plt.hist().**

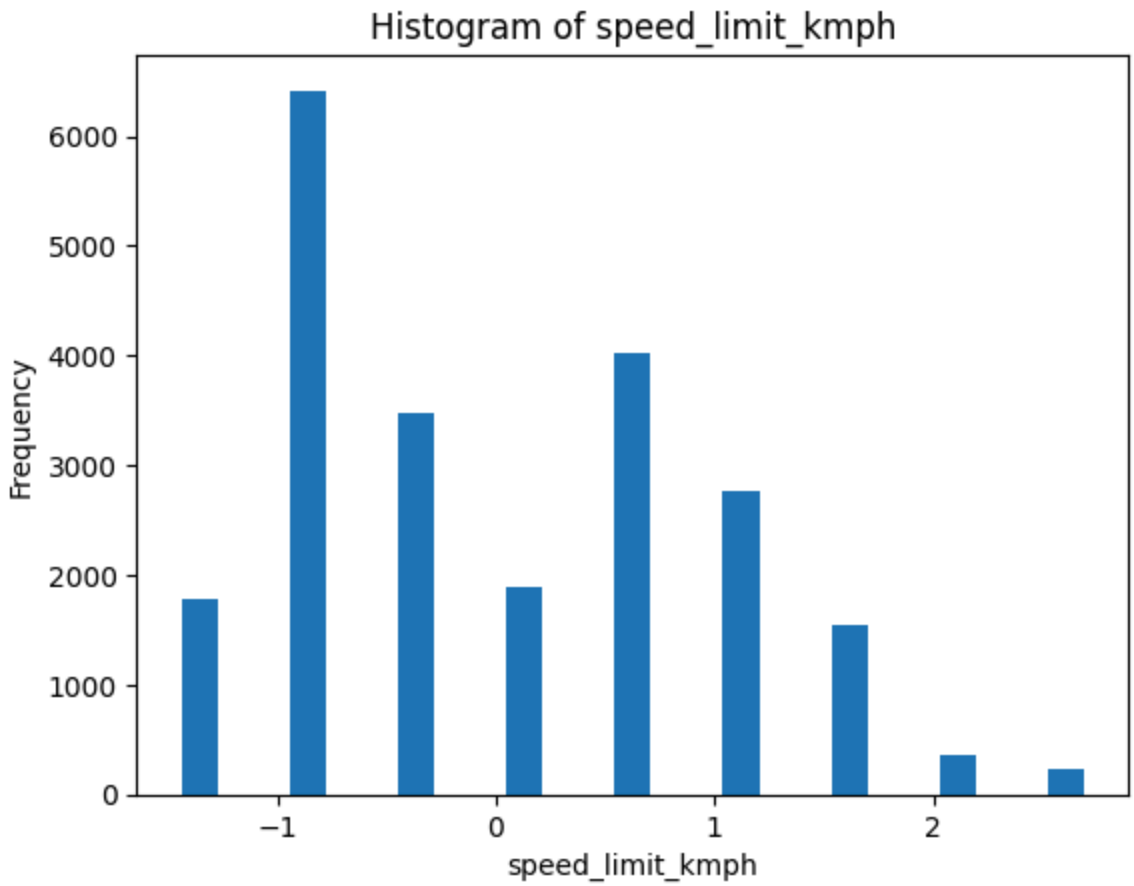
Insights:

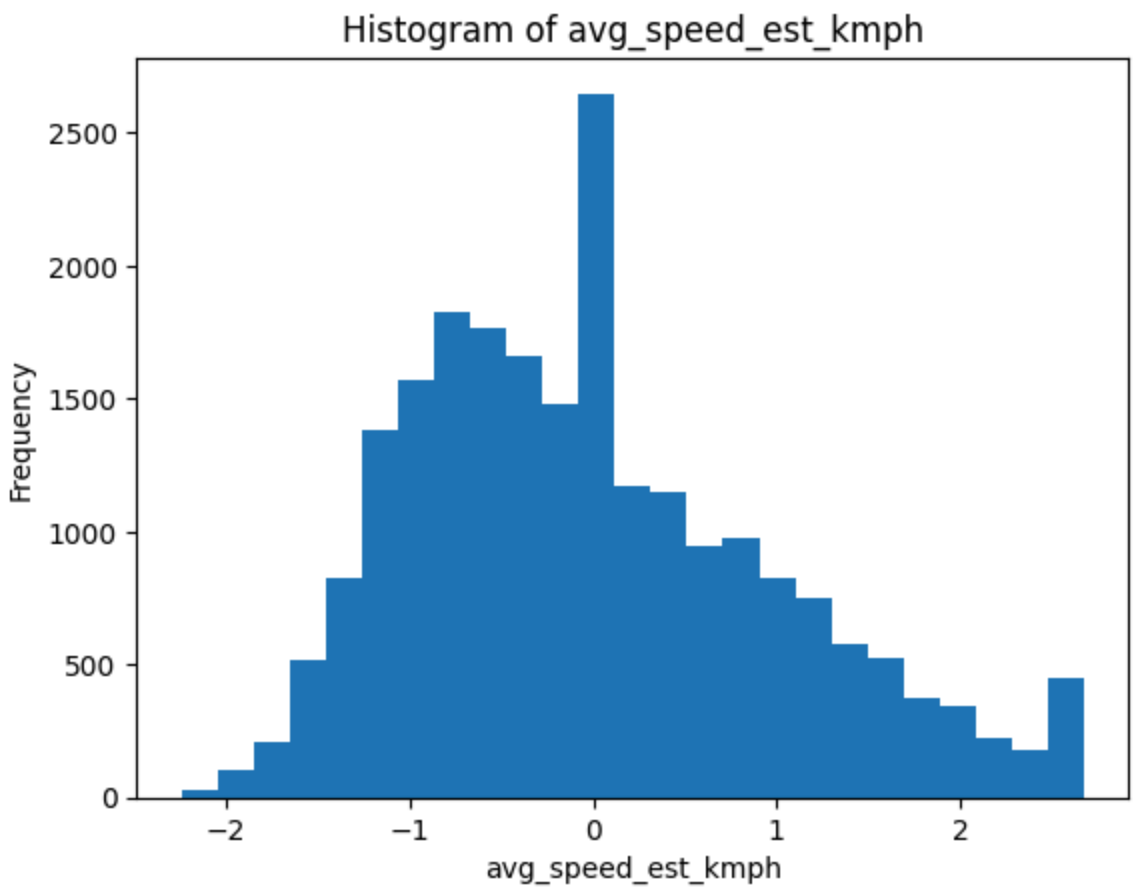
- Speed-related variables show multimodal distributions, reflecting different road types.
- Casualty and injury variables are right-skewed, indicating many minor accidents and few severe ones.
- Damage cost shows a heavy-tailed distribution, implying rare but extremely costly accidents.

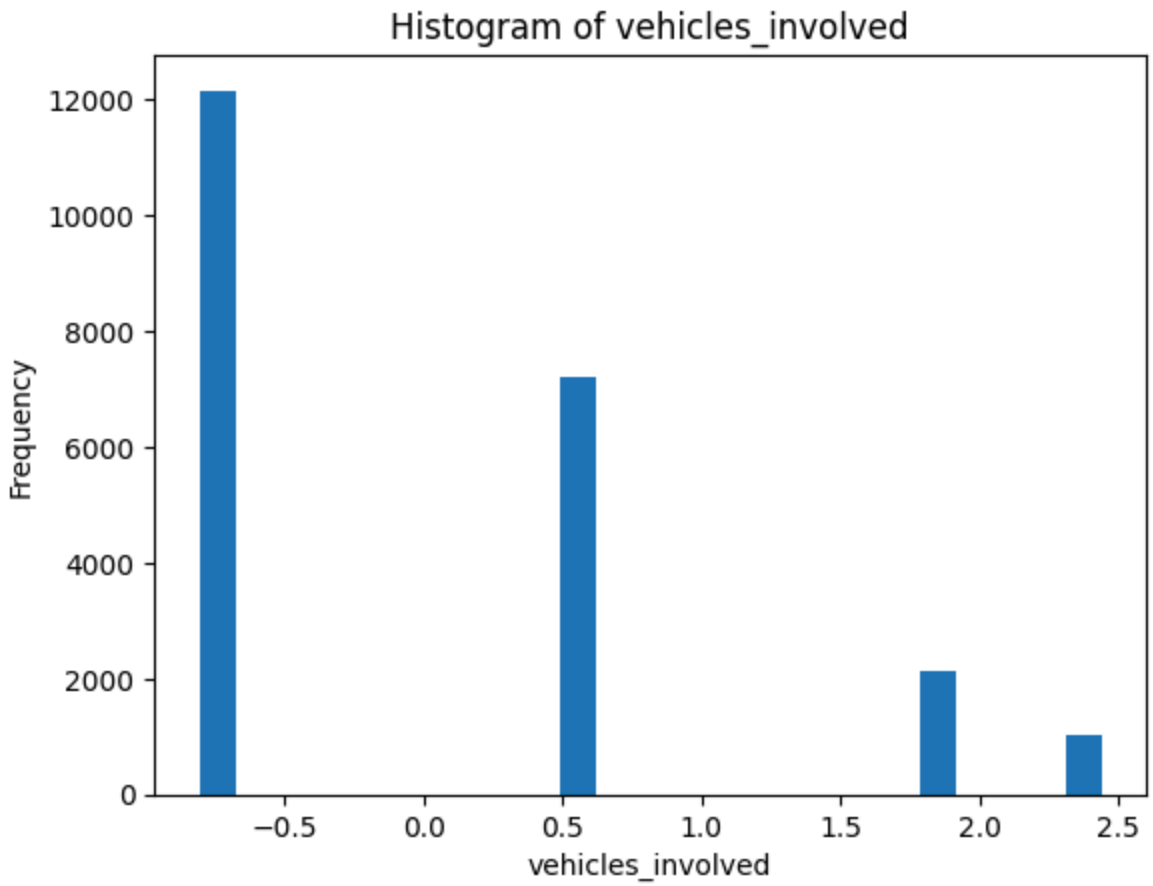
Why histograms:

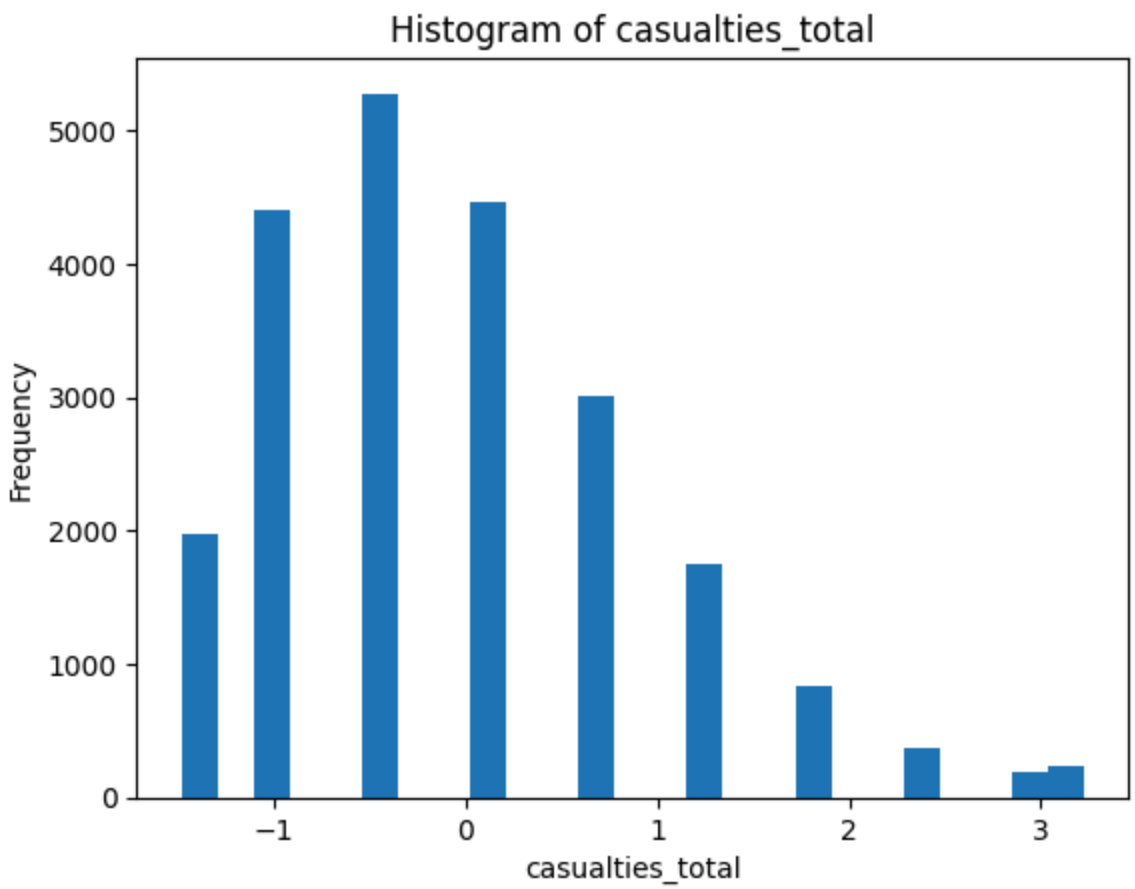
- They reveal distribution shape, skewness, and spread.
- Ideal for identifying non-normality.

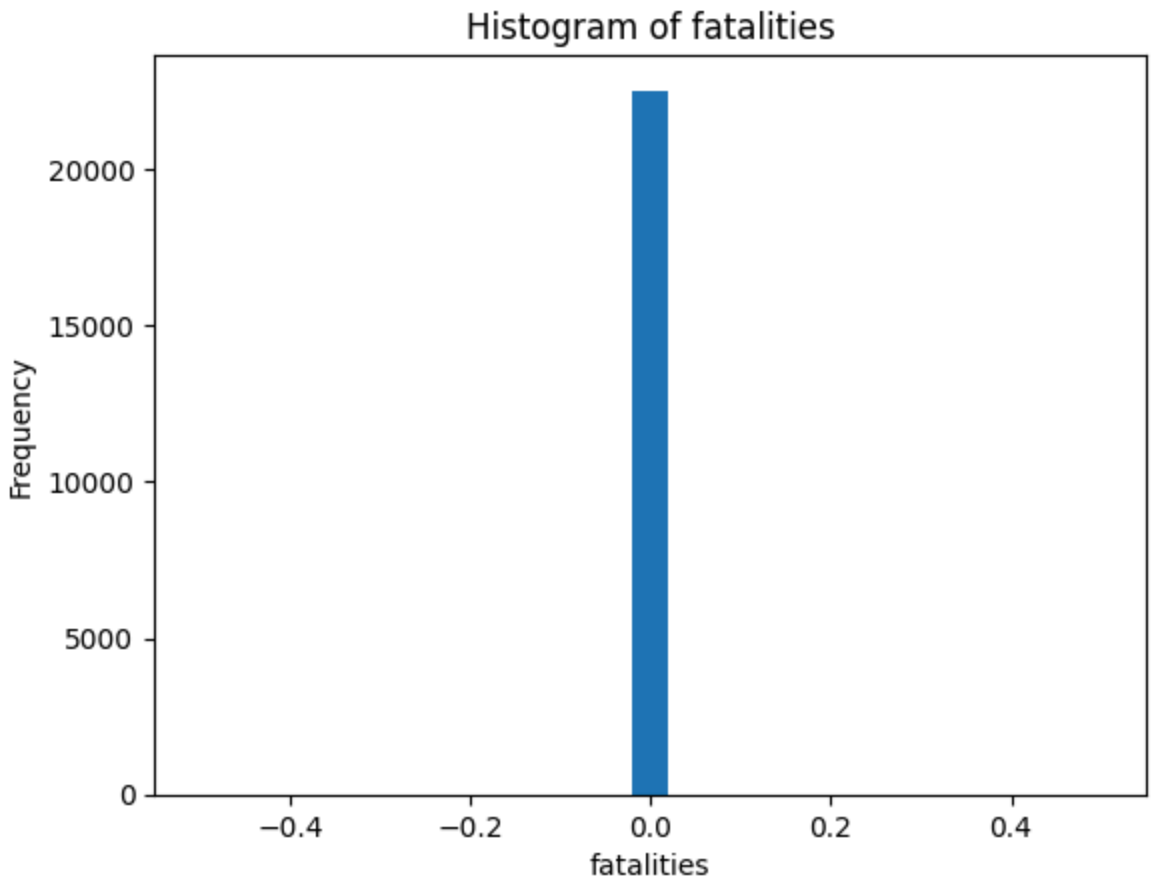
In [18]:
```
for col in num_cols:
    plt.hist(df[col], bins=25)
    plt.title(f"Histogram of {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
```

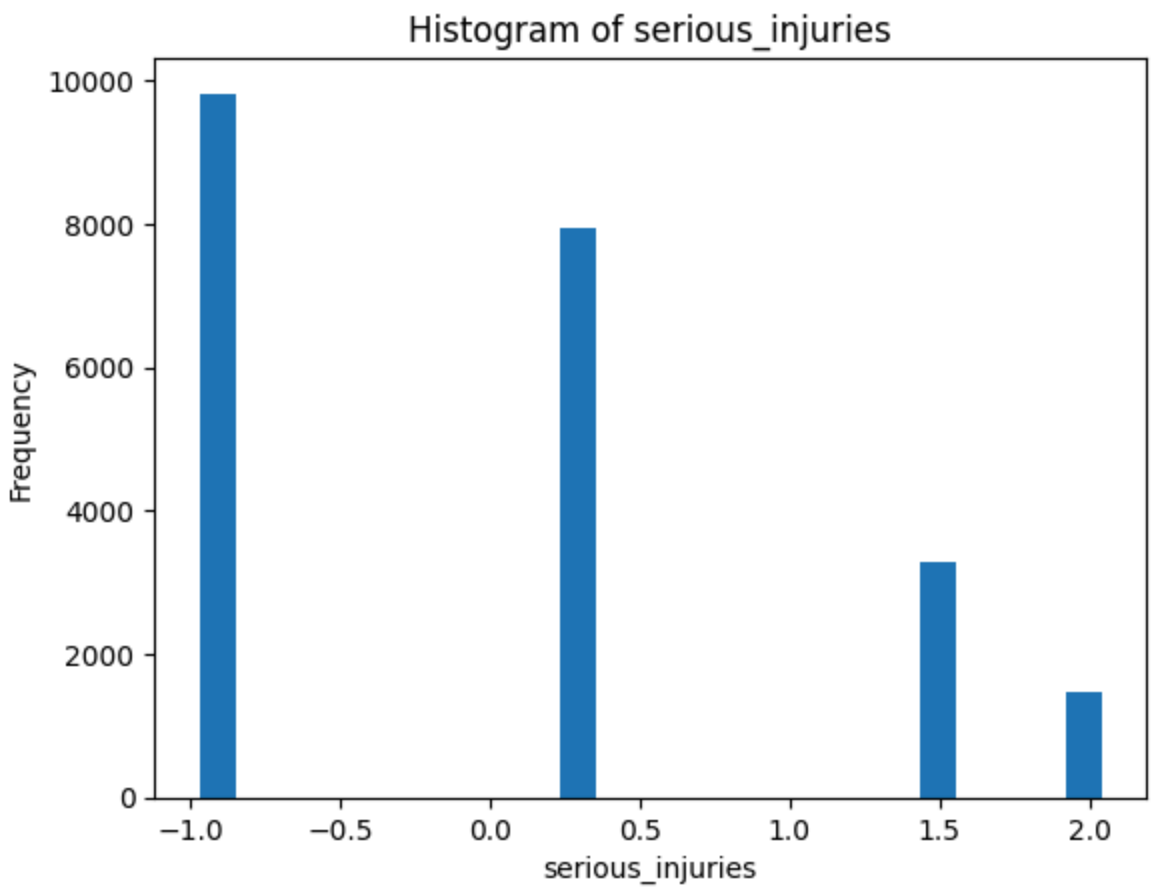

Histogram of speed_limit_kmph
Frequency
speed_limit_kmph
0
1000
2000
3000
4000
5000
6000
−1
0
1
2

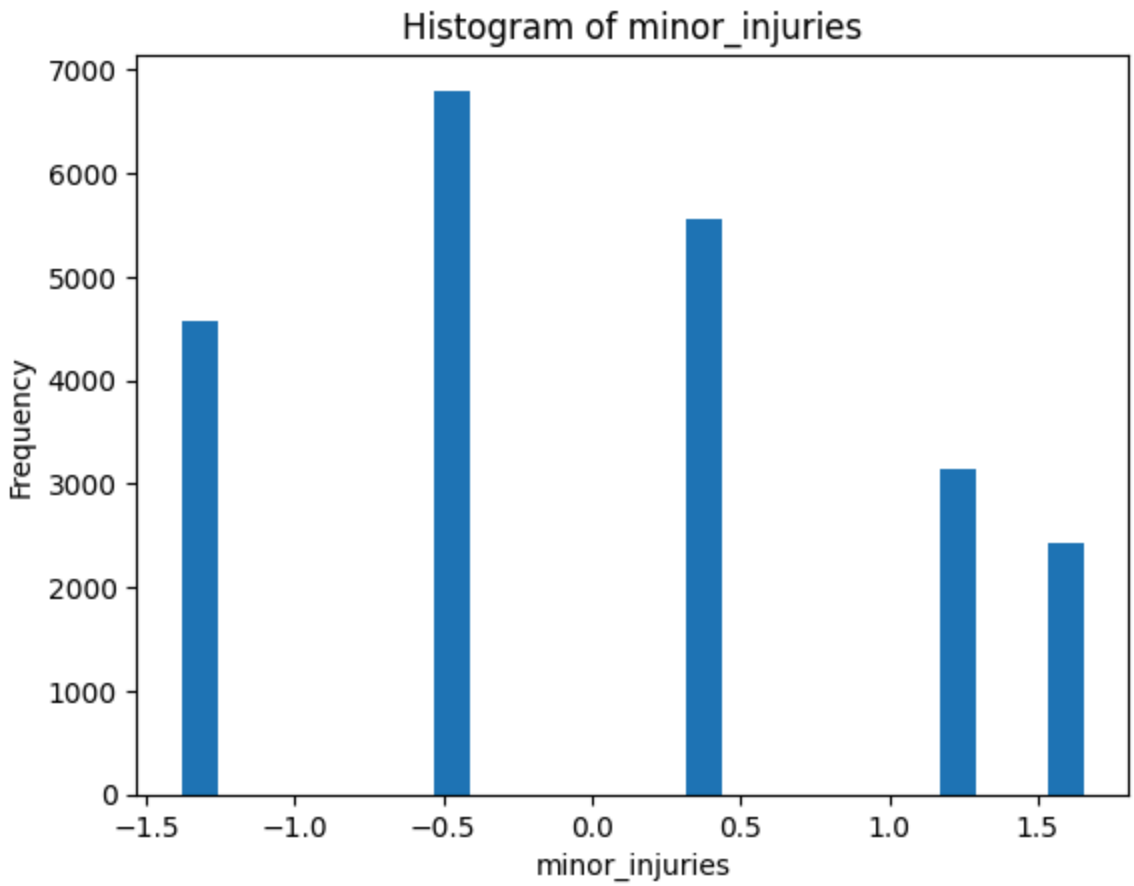

Histogram of avg_speed_est_kmph
Frequency
avg_speed_est_kmph
0
500
1000
1500
2000
2500
−2
−1
0
1
2

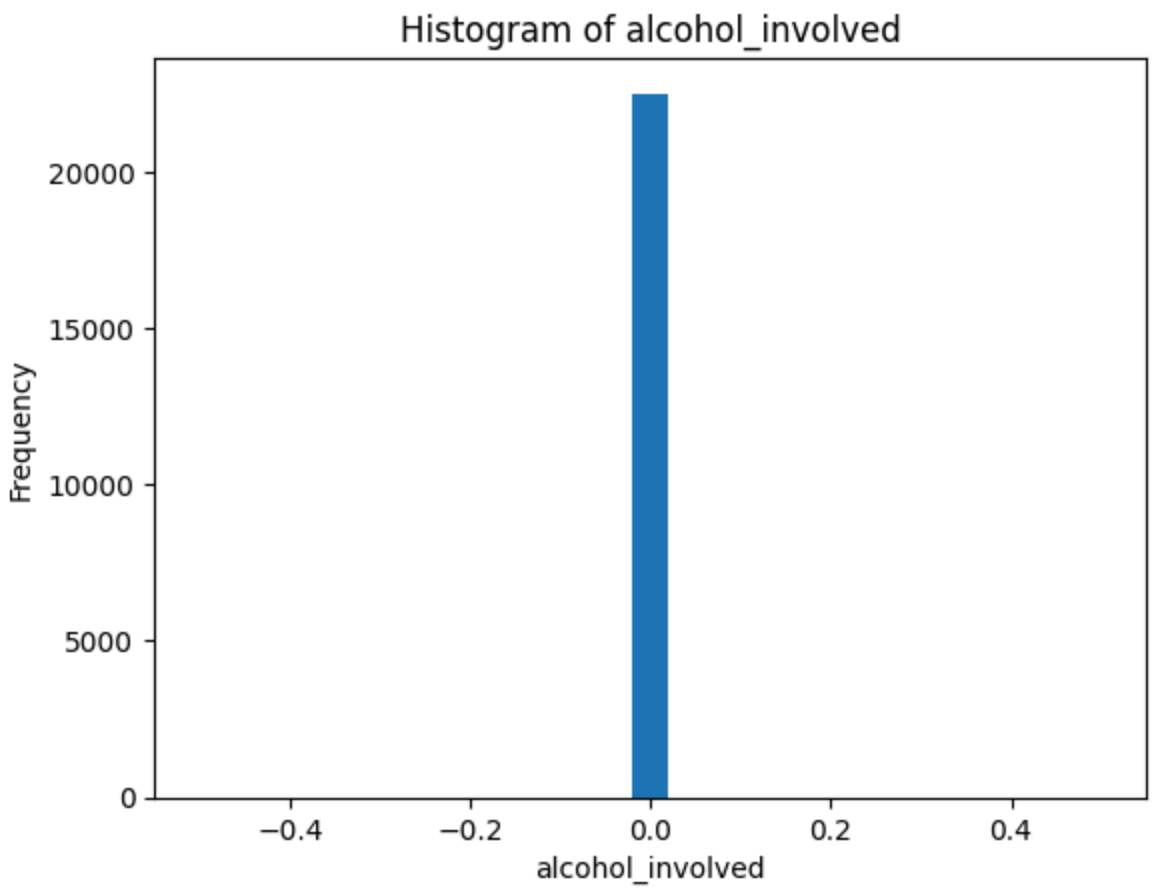

Histogram of vehicles_involved
Frequency
vehicles_involved
0
2000
4000
6000
8000
10000
12000
−0.5
0.0
0.5
1.0
1.5
2.0
2.5

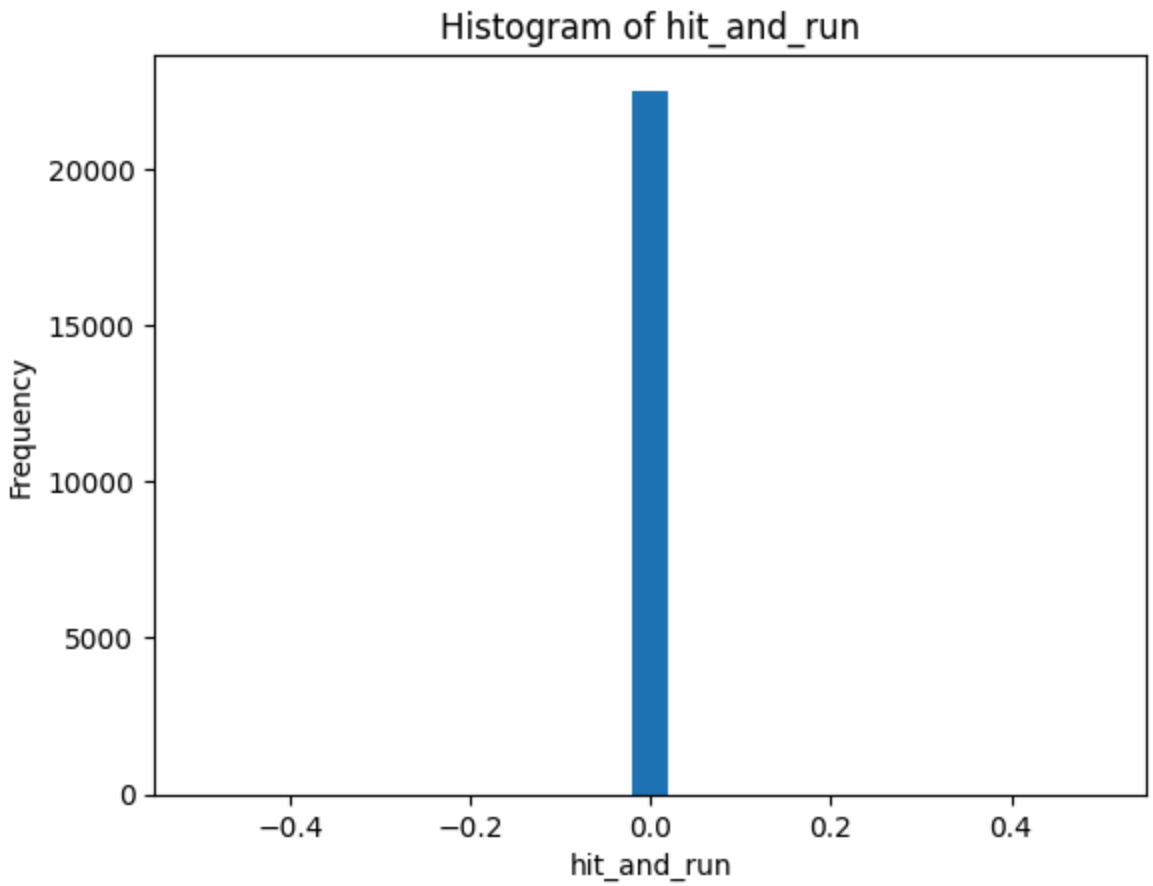

Histogram of casualties_total
Frequency
casualties_total
0
1000
2000
3000
4000
5000
−1
0
1
2
3

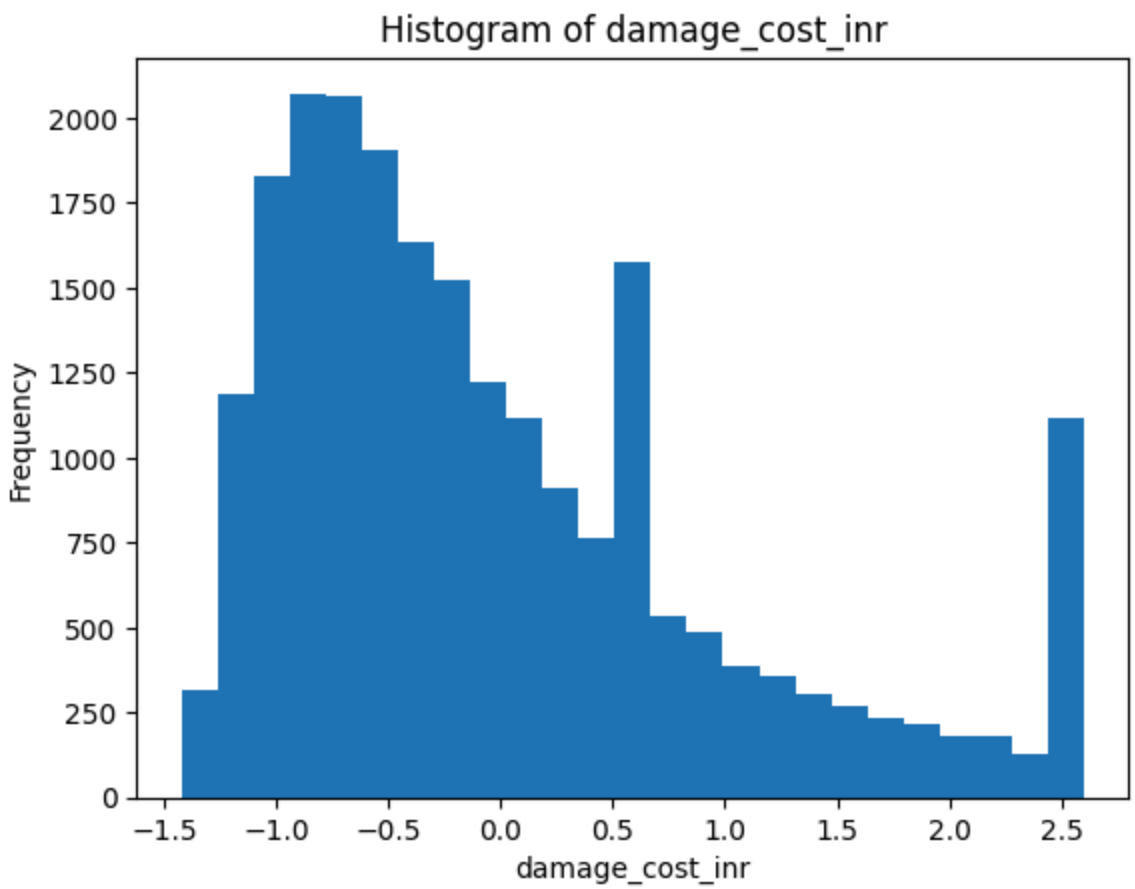

Histogram of fatalities
Frequency
20000
15000
10000
5000
0
−0.4
−0.2
0.0
0.2
0.4
fatalities


Histogram of serious_injuries
Frequency
10000
8000
6000
4000
2000
0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
serious_injuries


Histogram of minor_injuries
Frequency
minor_injuries
7000
6000
5000
4000
3000
2000
1000
0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


Histogram of alcohol_involved
Frequency
alcohol_involved
20000
15000
10000
5000
0
−0.4
−0.2
0.0
0.2
0.4


Histogram of hit_and_run
Frequency
20000
15000
10000
5000
0
−0.4
−0.2
0.0
0.2
0.4
hit_and_run


Histogram of damage_cost_inr
Frequency
2000
1750
1500
1250
1000
750
500
250
0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
damage_cost_inr

KDE Plots

Kernel Density Estimation plots were created using **sns.kdeplot().**

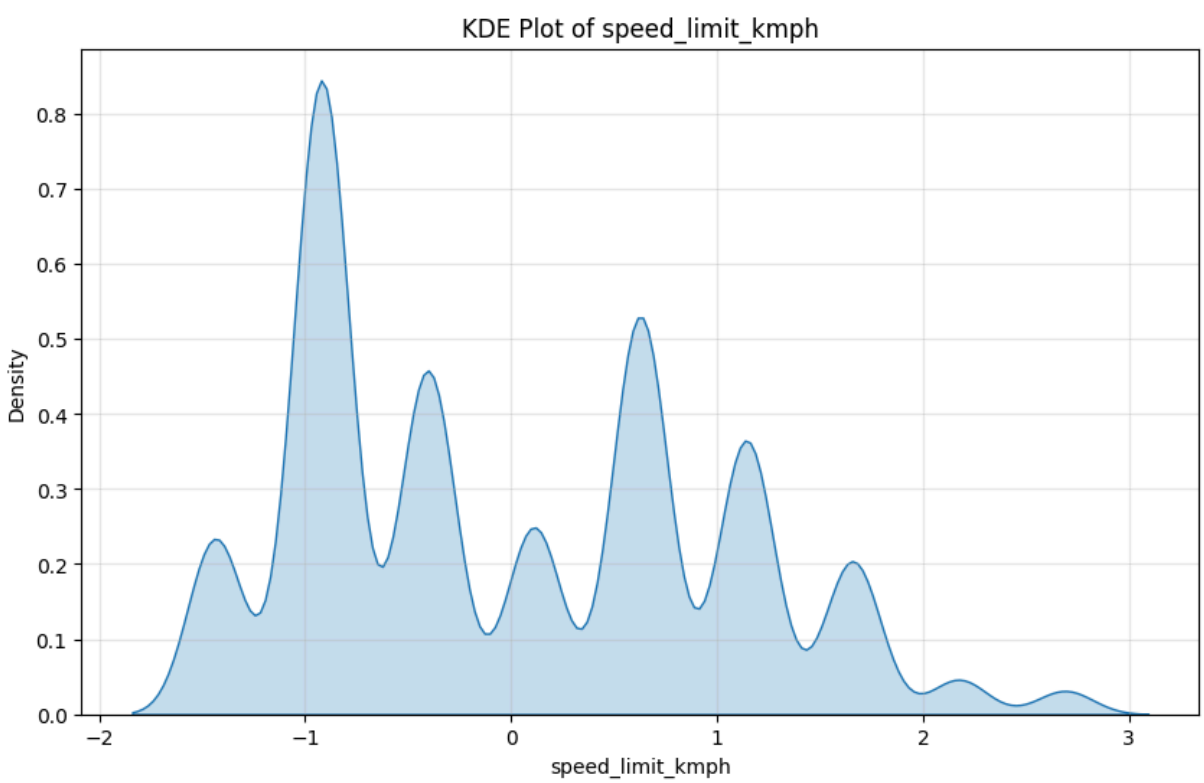
Why KDE:

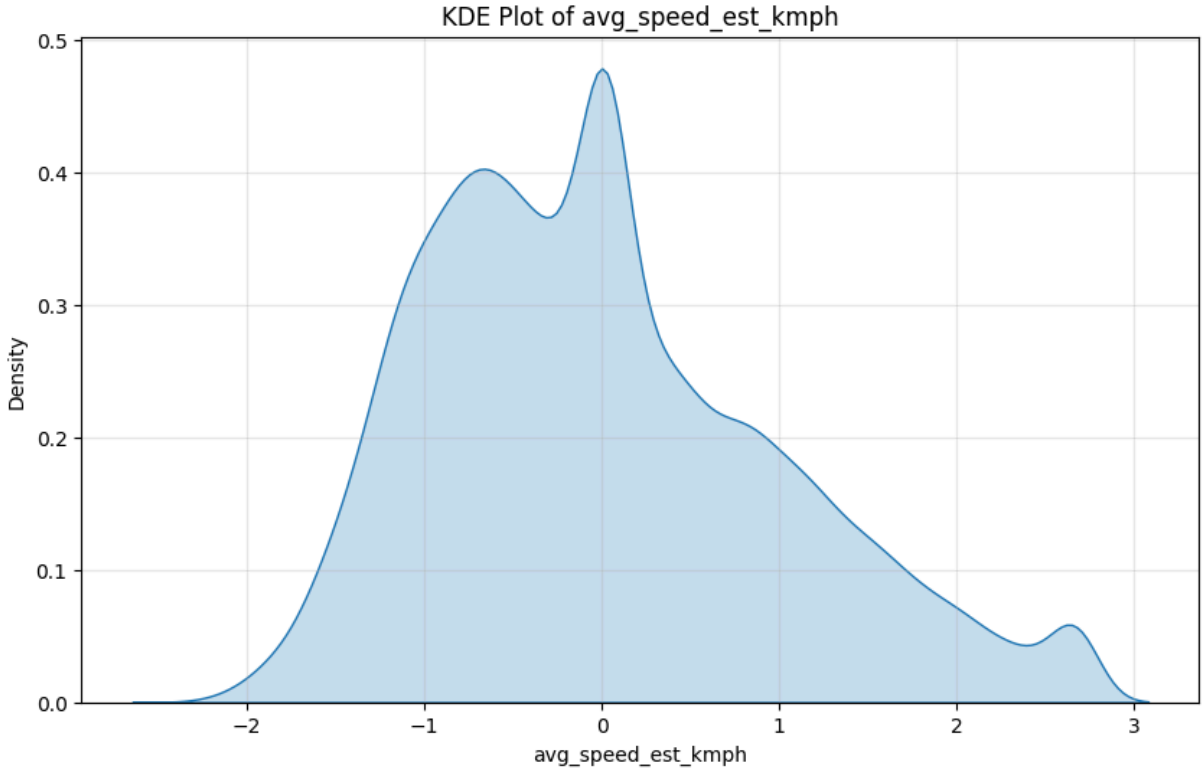
- KDE provides a smooth estimate of the probability density function.
- Useful when histograms appear noisy.

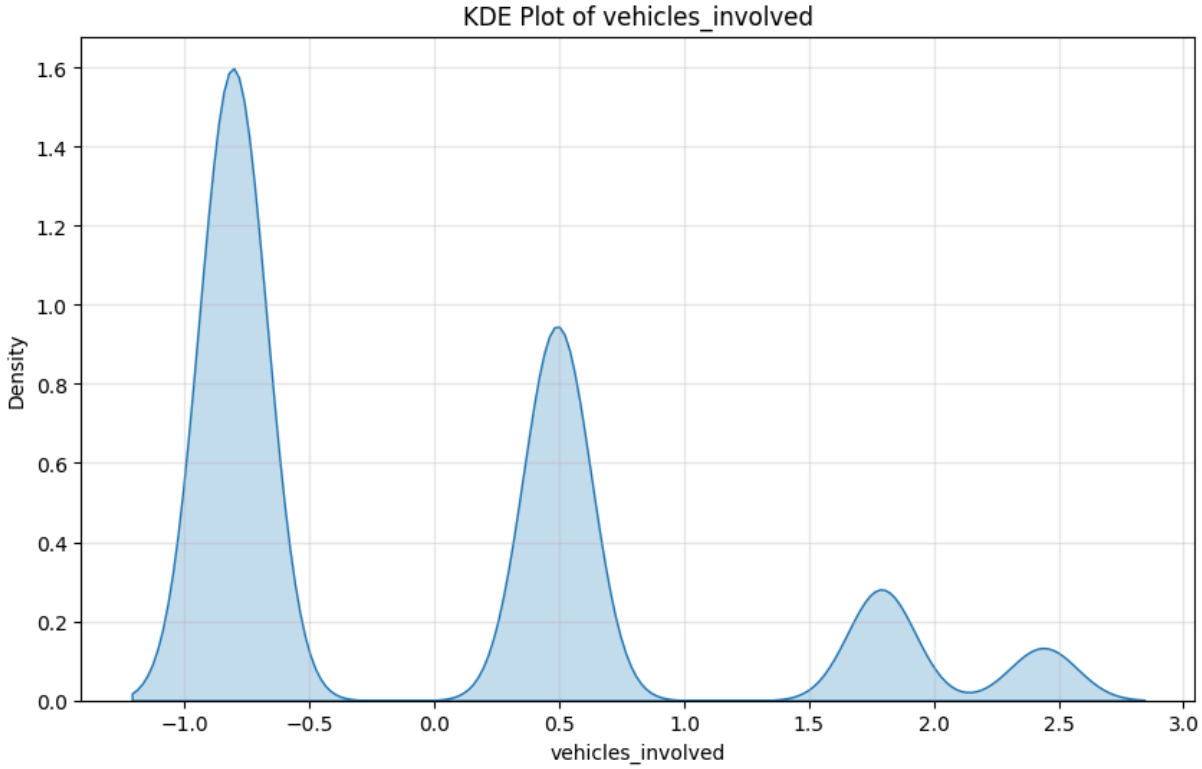
Insights:

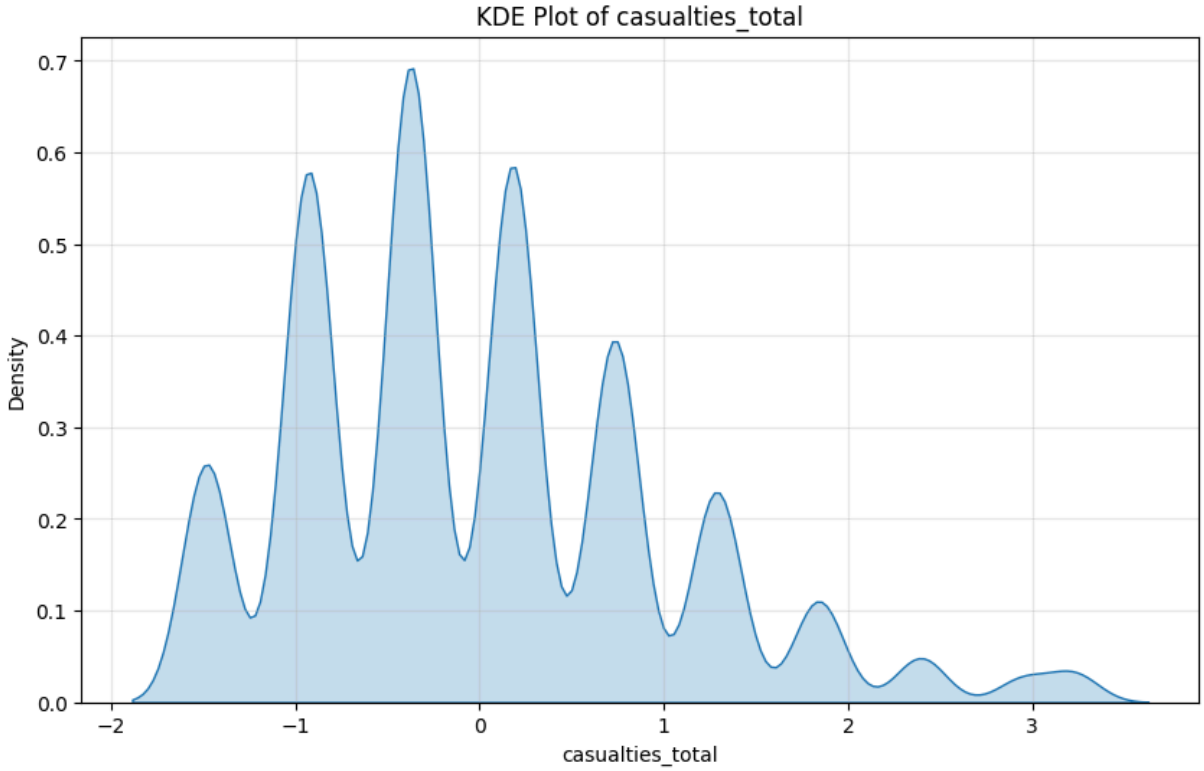
- Speed and casualty distributions show multiple density peaks, suggesting heterogeneous accident scenarios.
- Damage cost KDE confirms strong positive skewness.

In [47]:
```
for col in num_cols:
    if df[col].nunique() > 1 and df[col].std() > 0:
        plt.figure(figsize=(10, 6))
        sns.kdeplot(df[col], fill=True, warn_singular=False)
        plt.title(f"KDE Plot of {col}")
        plt.xlabel(col)
        plt.ylabel('Density')
        plt.grid(alpha=0.3)
        plt.show()
    else:
        print(f"Skipping {col}: No variance (all values are the same)")
```

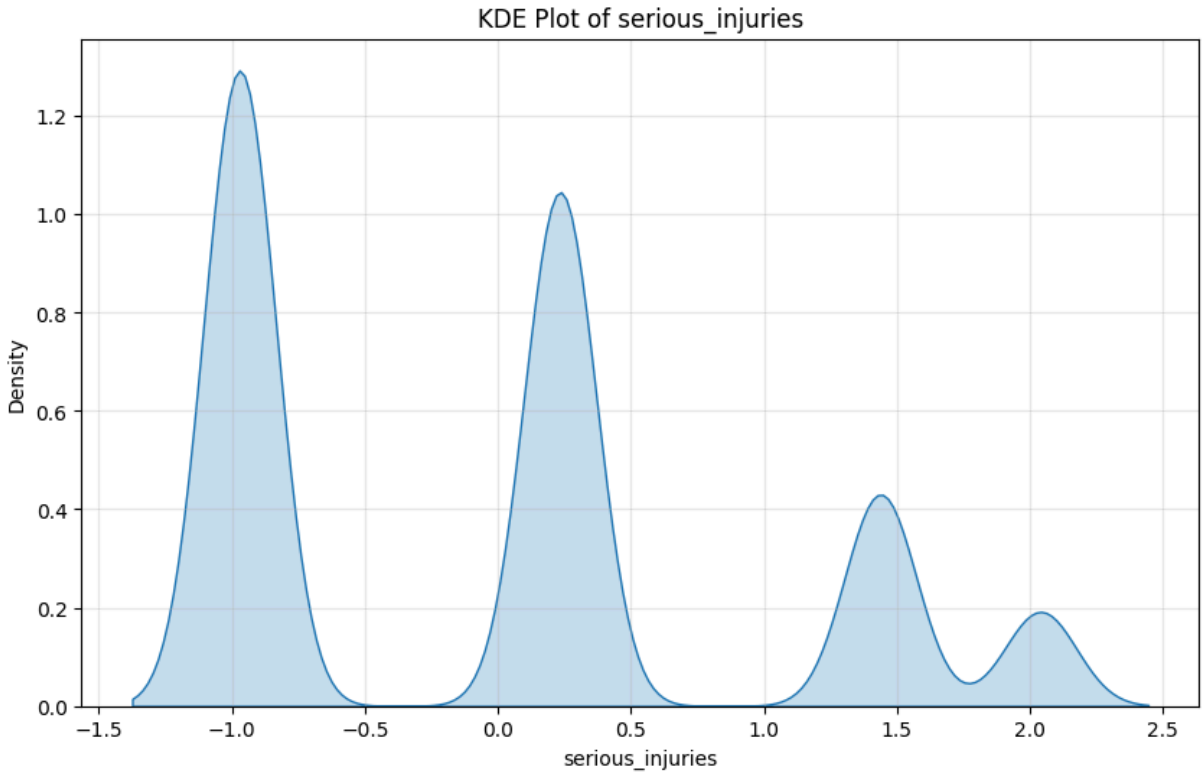


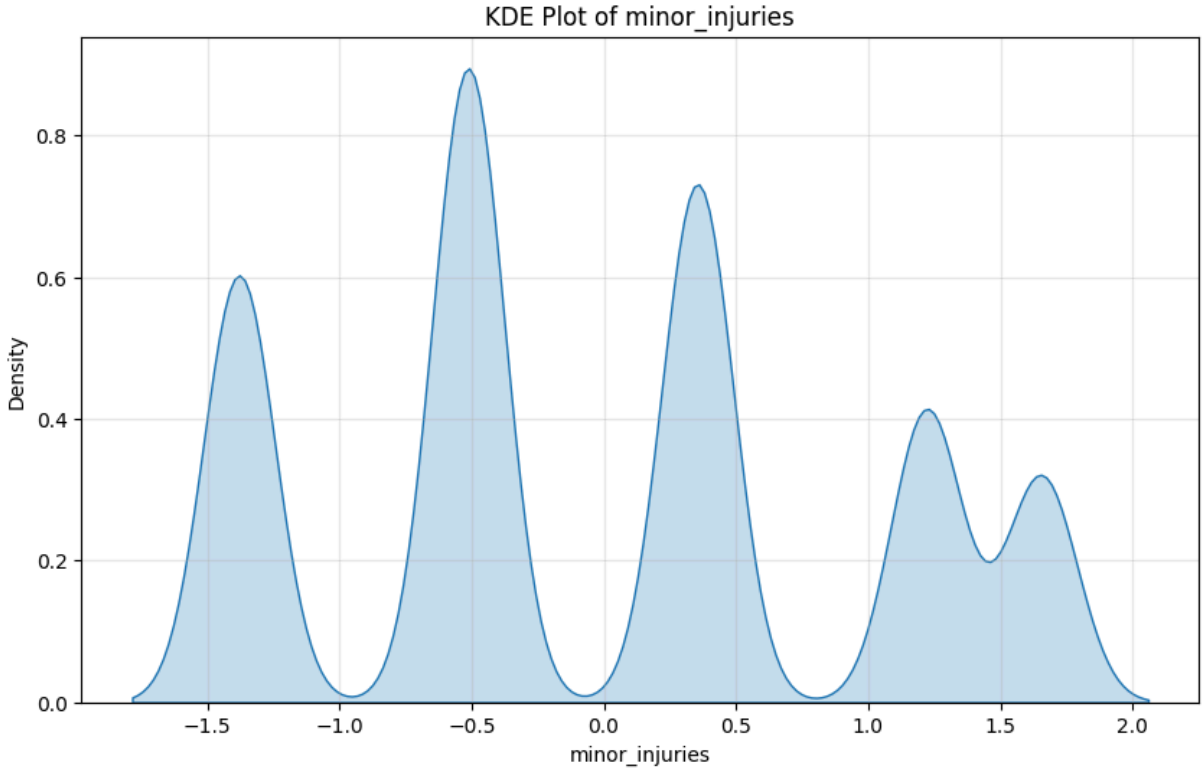

KDE Plot of avg_speed_est_kmph
Density
avg_speed_est_kmph
0.0
0.1
0.2
0.3
0.4
0.5
−2
−1
0
1
2
3


KDE Plot of vehicles_involved
Density
vehicles_involved
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

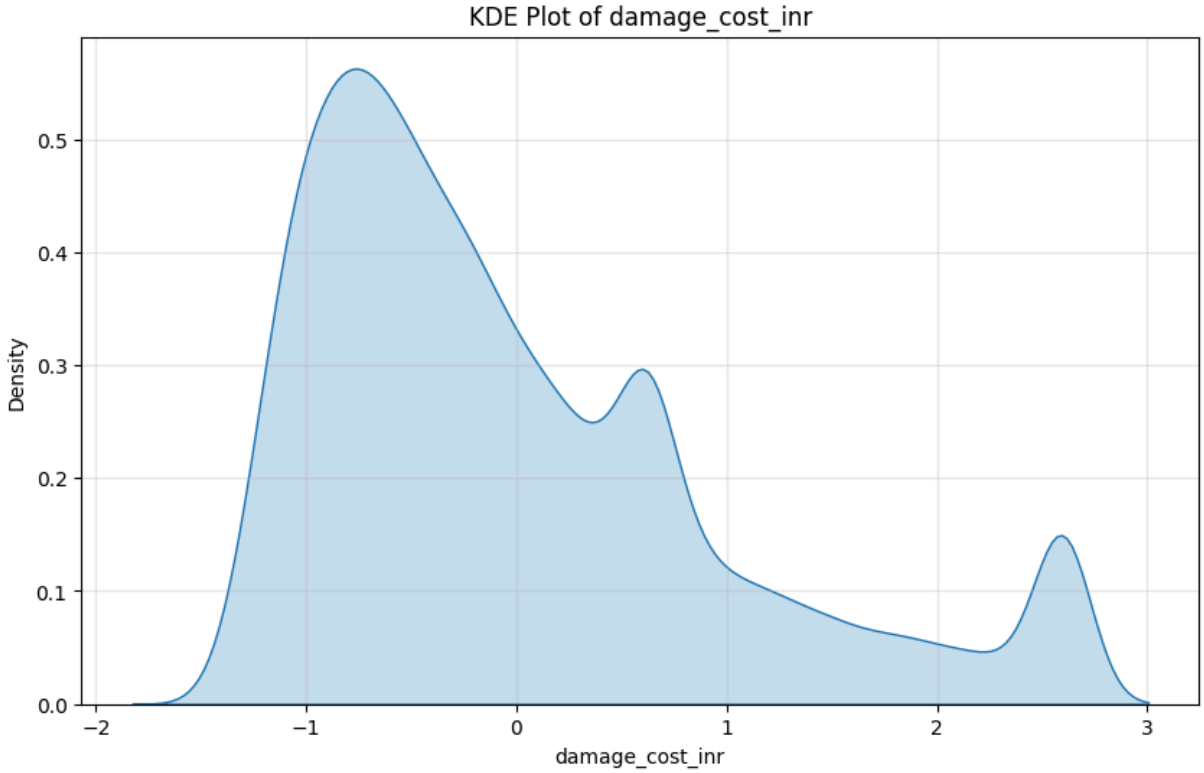



```
Skipping fatalities: No variance (all values are the same)
```






```
Skipping alcohol_involved: No variance (all values are the same)
Skipping hit_and_run: No variance (all values are the same)
```



Violin Plots

Violin plots were used for selected numerical columns.

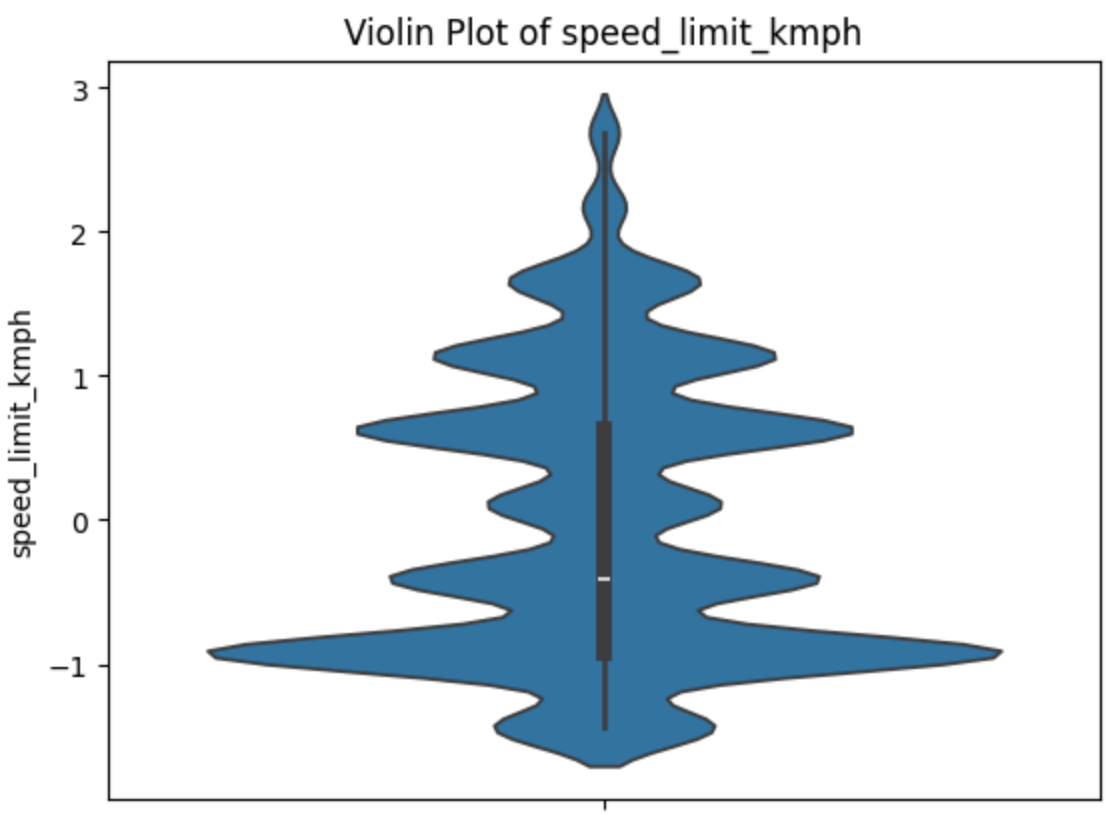
Why violin plots:

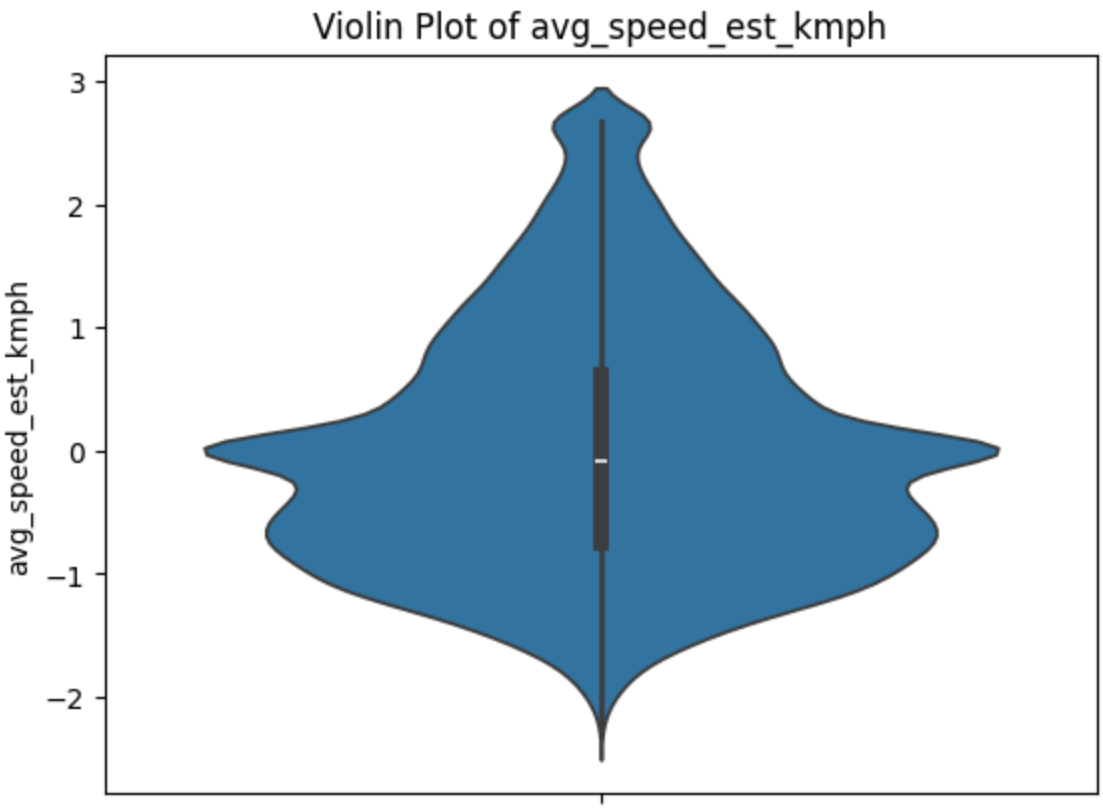
- Combine boxplot summary with full distribution shape.
- Effective for understanding density concentration.

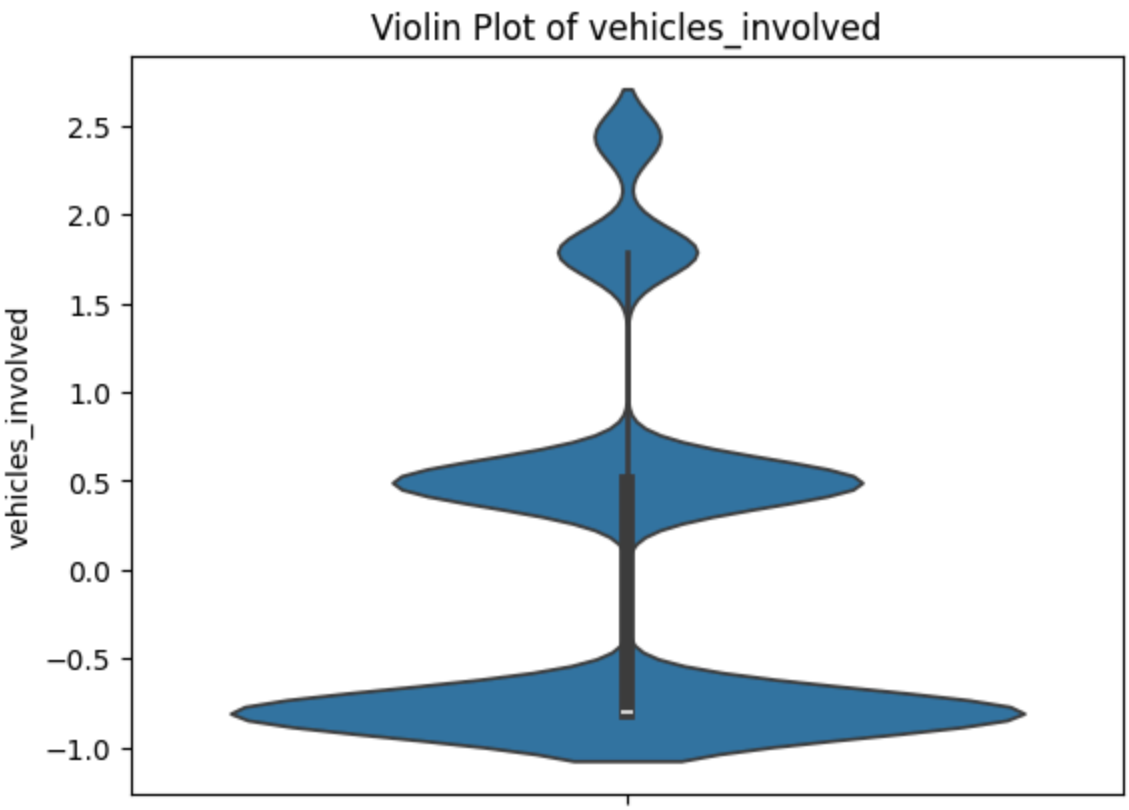
Insights:

- Injury-related variables show dense lower ranges with long upper tails.
- Indicates most accidents cause few injuries, but some result in severe harm.

In [20]:
```
for col in num_cols[:3]:
    sns.violinplot(y=df[col])
    plt.title(f"Violin Plot of {col}")
    plt.show()
```







---

8. Bivariate Analysis

Scatter Plots

Scatter plots were plotted between numerical pairs using **sns.scatterplot().**

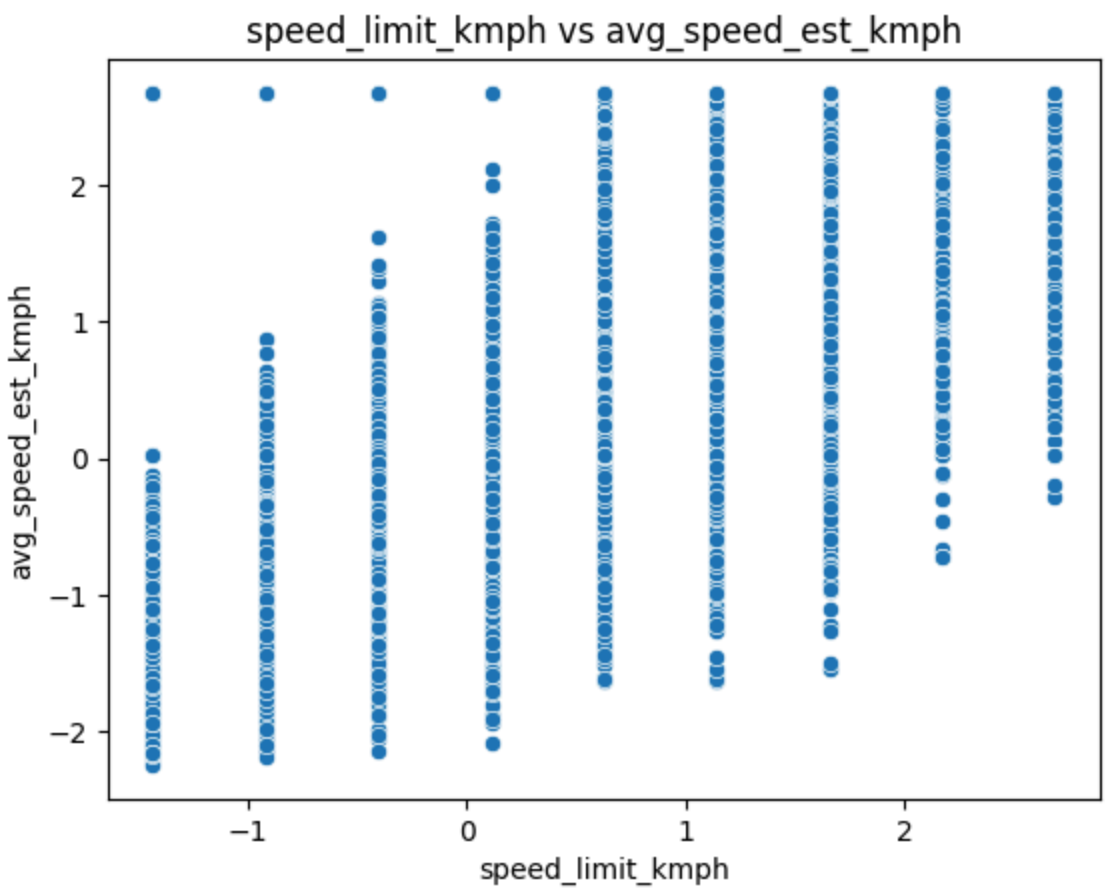
Insights:

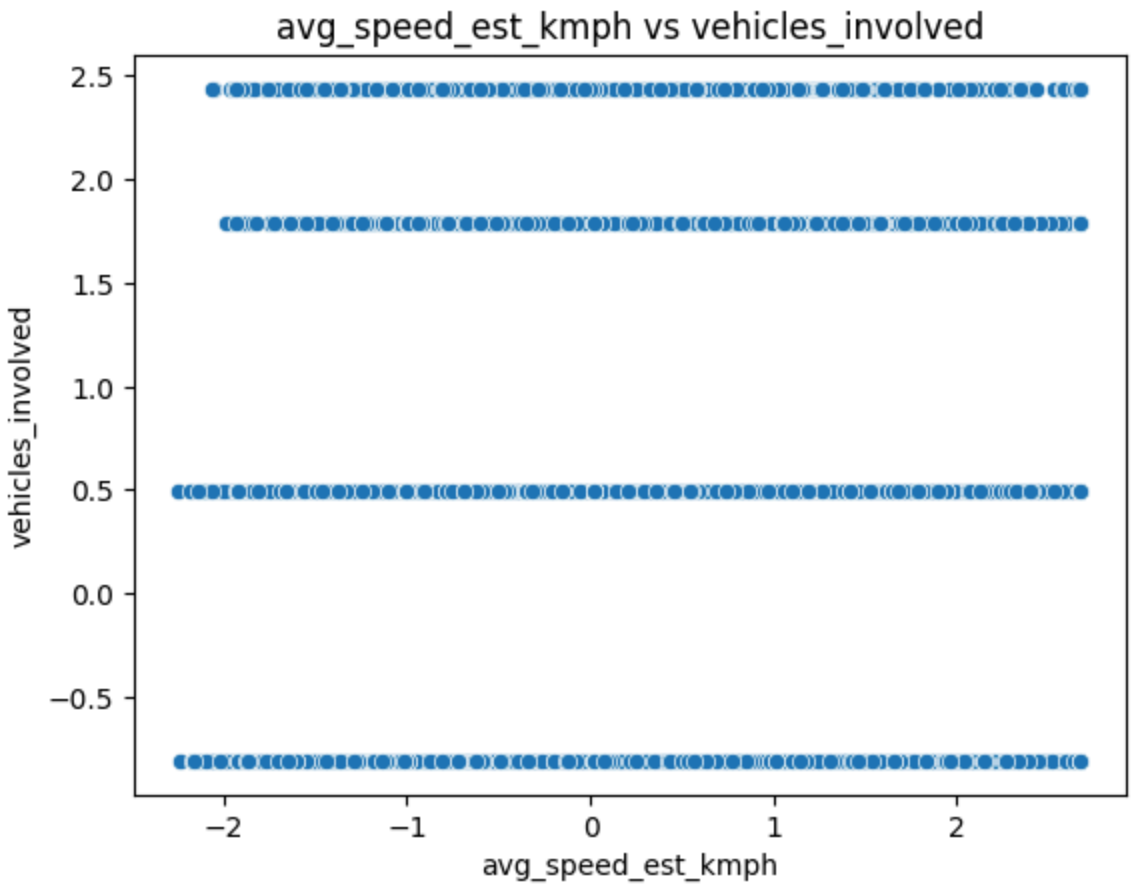
- Higher estimated speeds are weakly associated with higher casualties.
- Vehicle count correlates positively with total injuries.

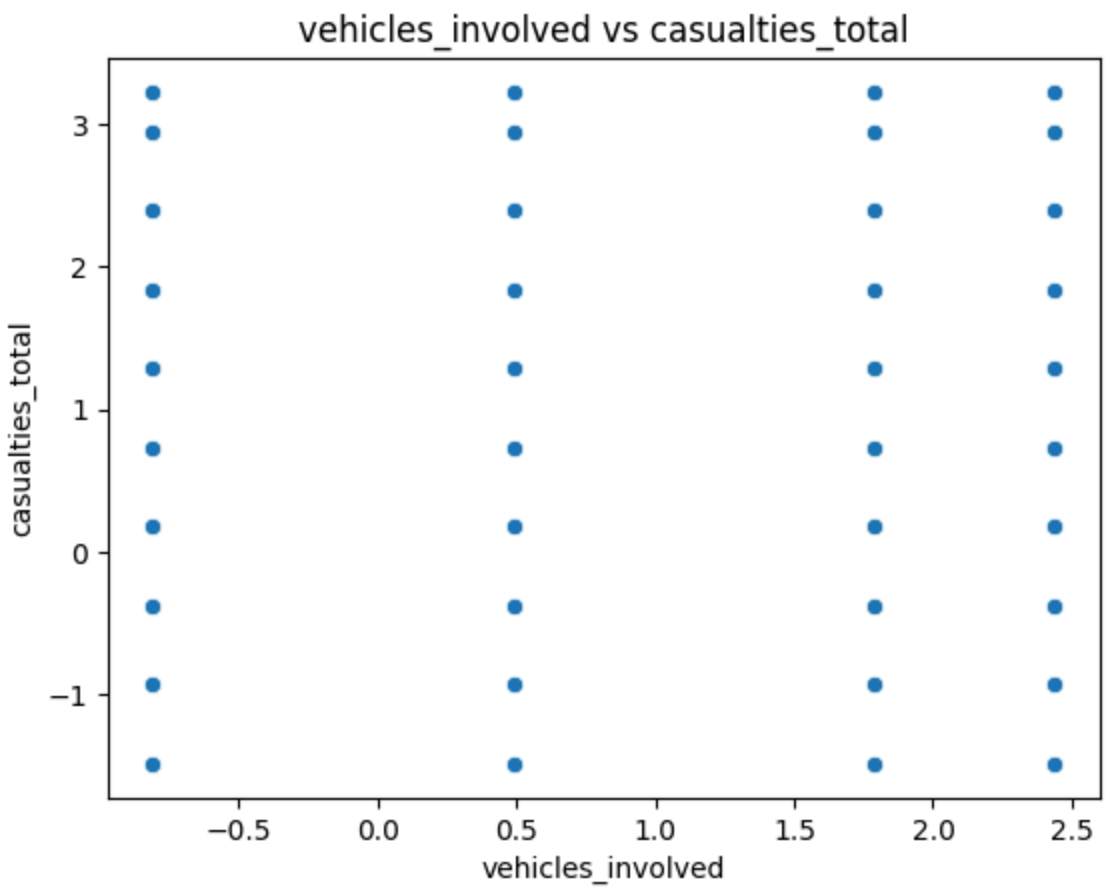
Why scatter plots:

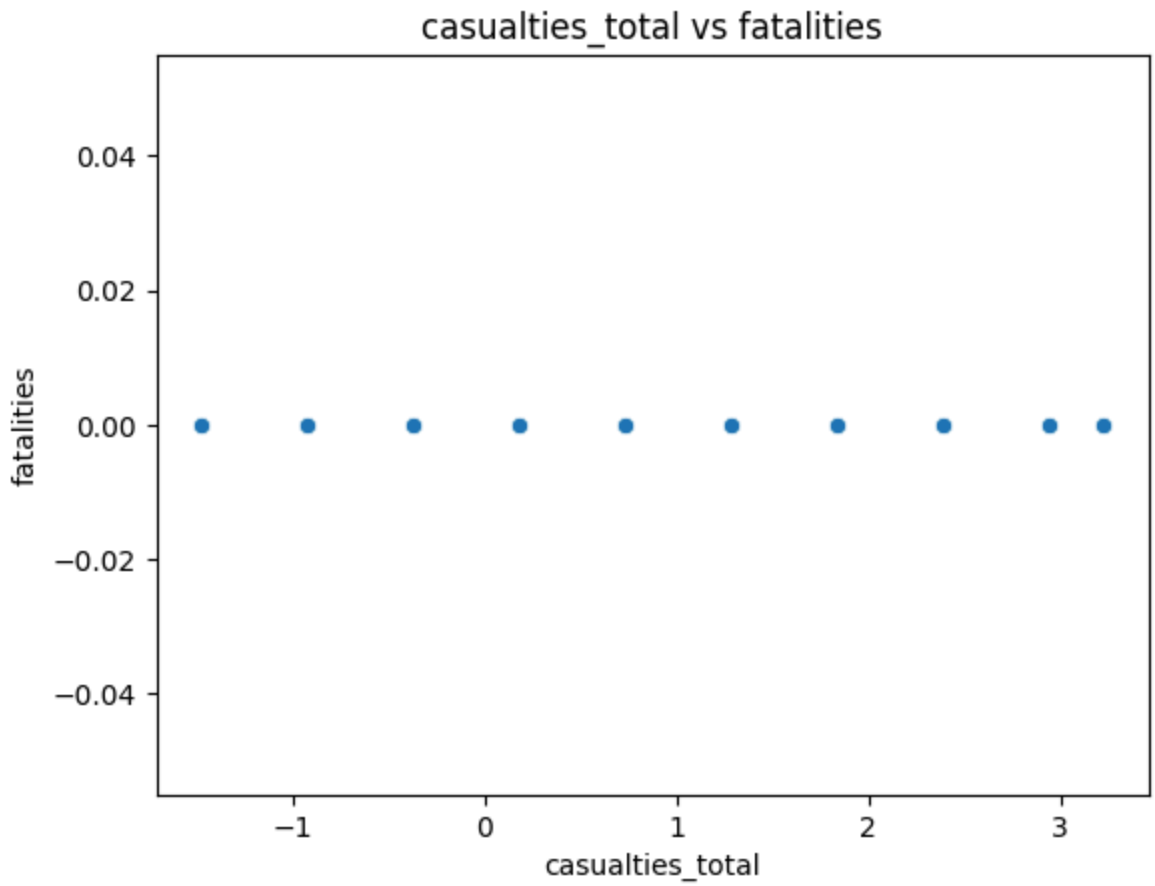
- Ideal for detecting linear or non-linear relationships.
- Helps identify clusters and trends.

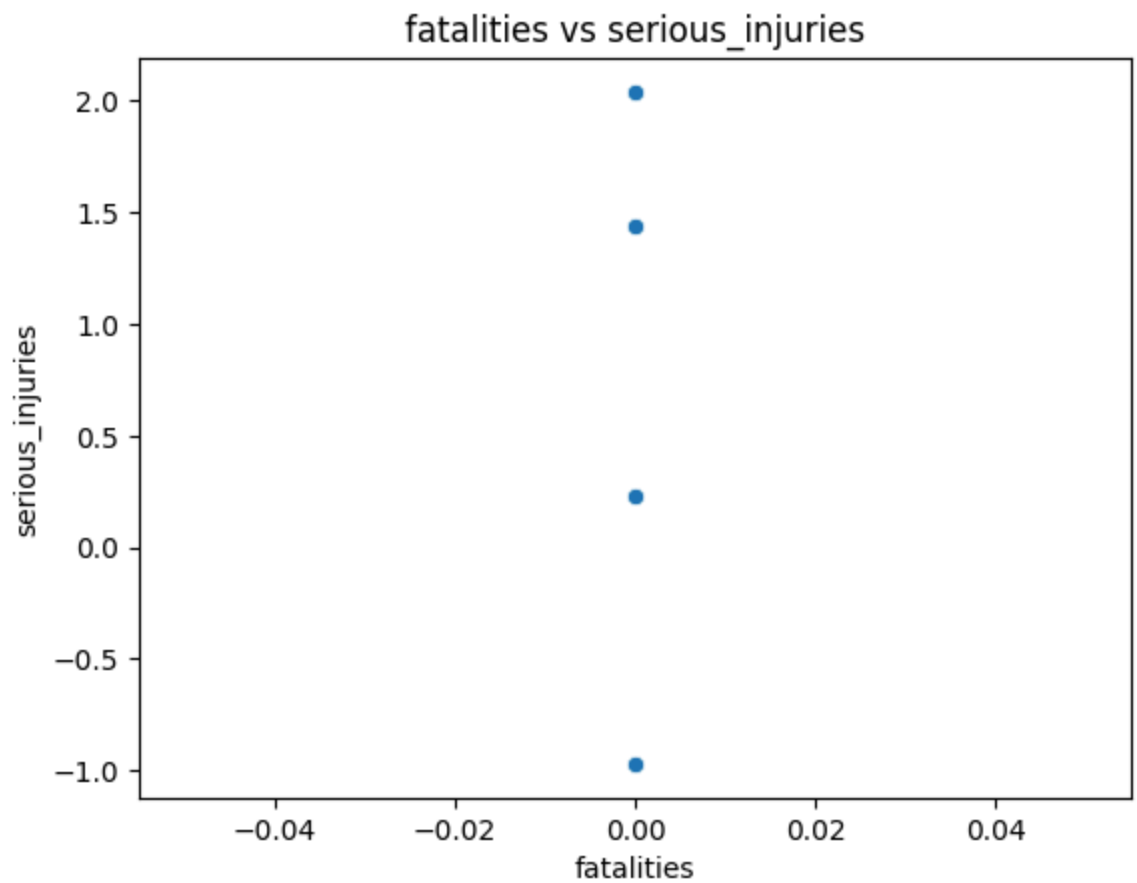
In [21]:
```
for i in range(len(num_cols)-1):
    sns.scatterplot(
        x=df[num_cols[i]],
        y=df[num_cols[i+1]]
    )
    plt.title(f"{num_cols[i]} vs {num_cols[i+1]}")
    plt.show()
```

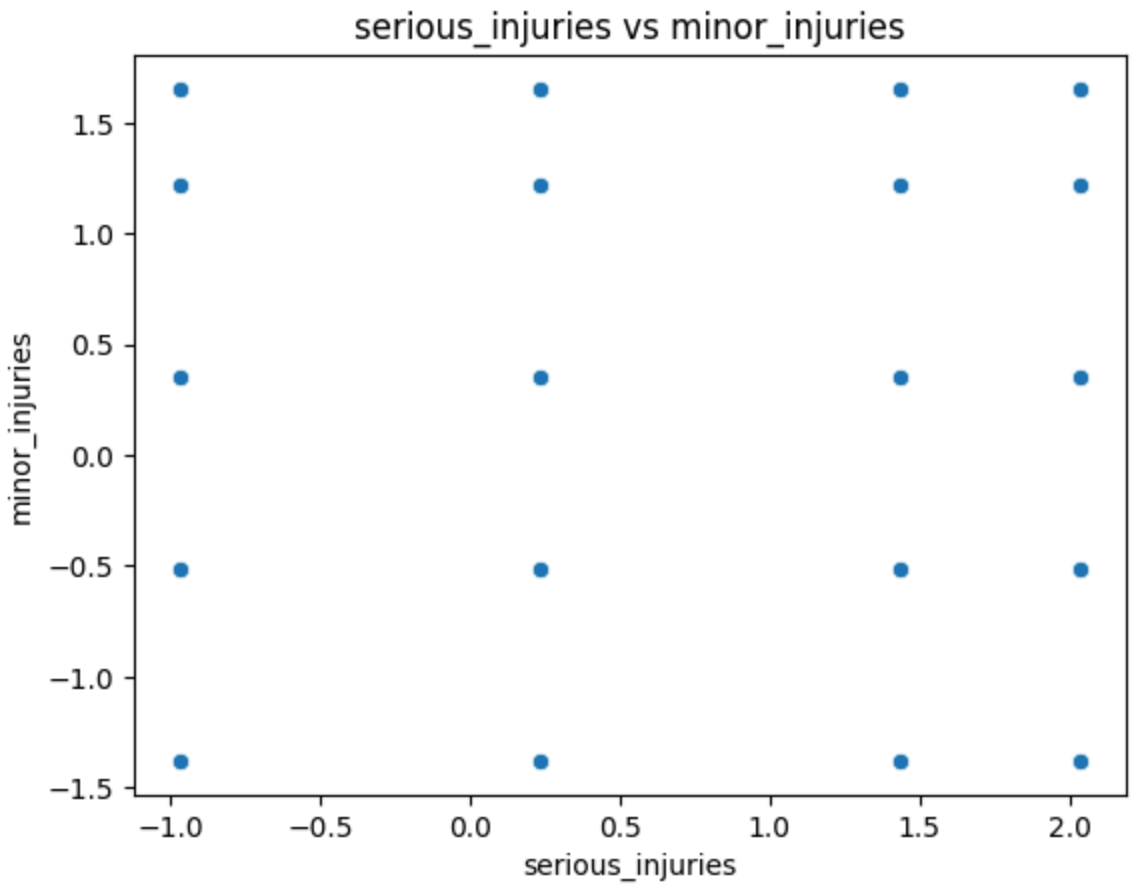


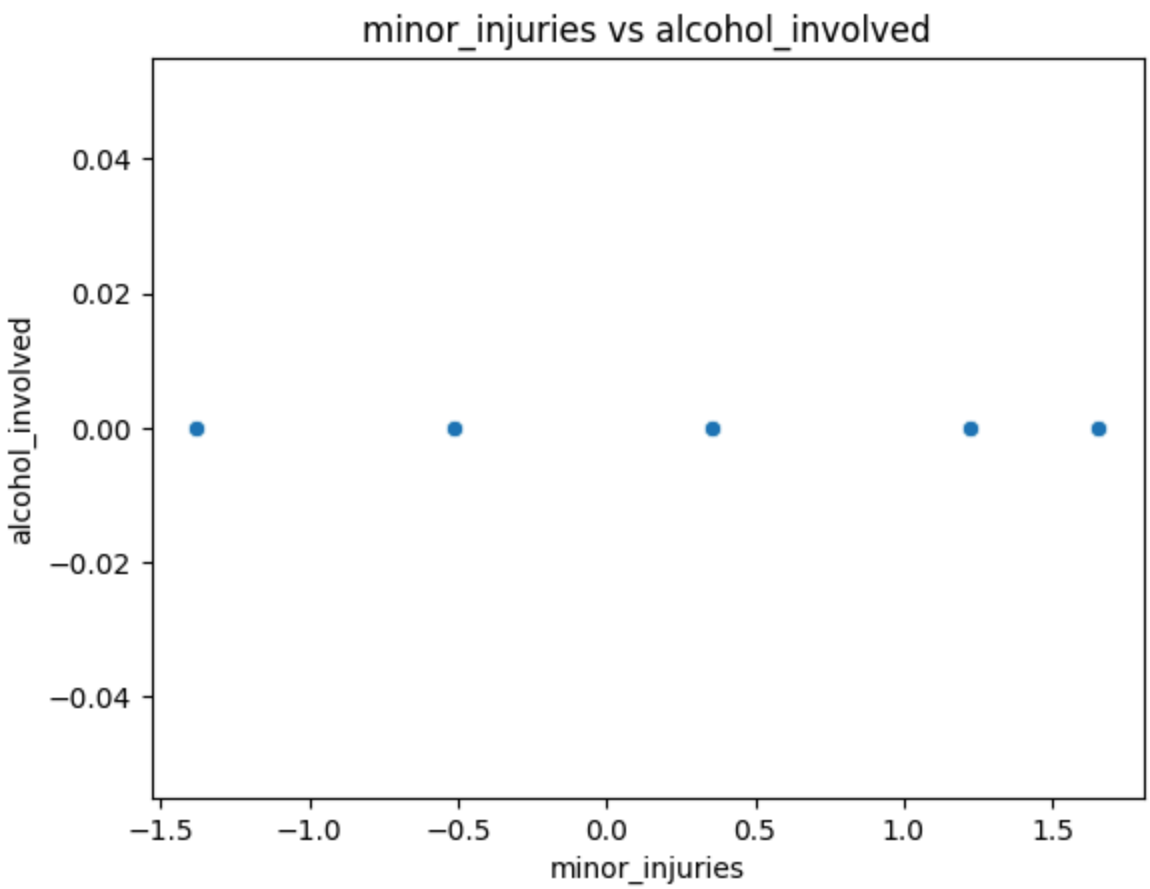

avg_speed_est_kmph vs vehicles_involved
vehicles_involved
avg_speed_est_kmph
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−2
−1
0
1
2

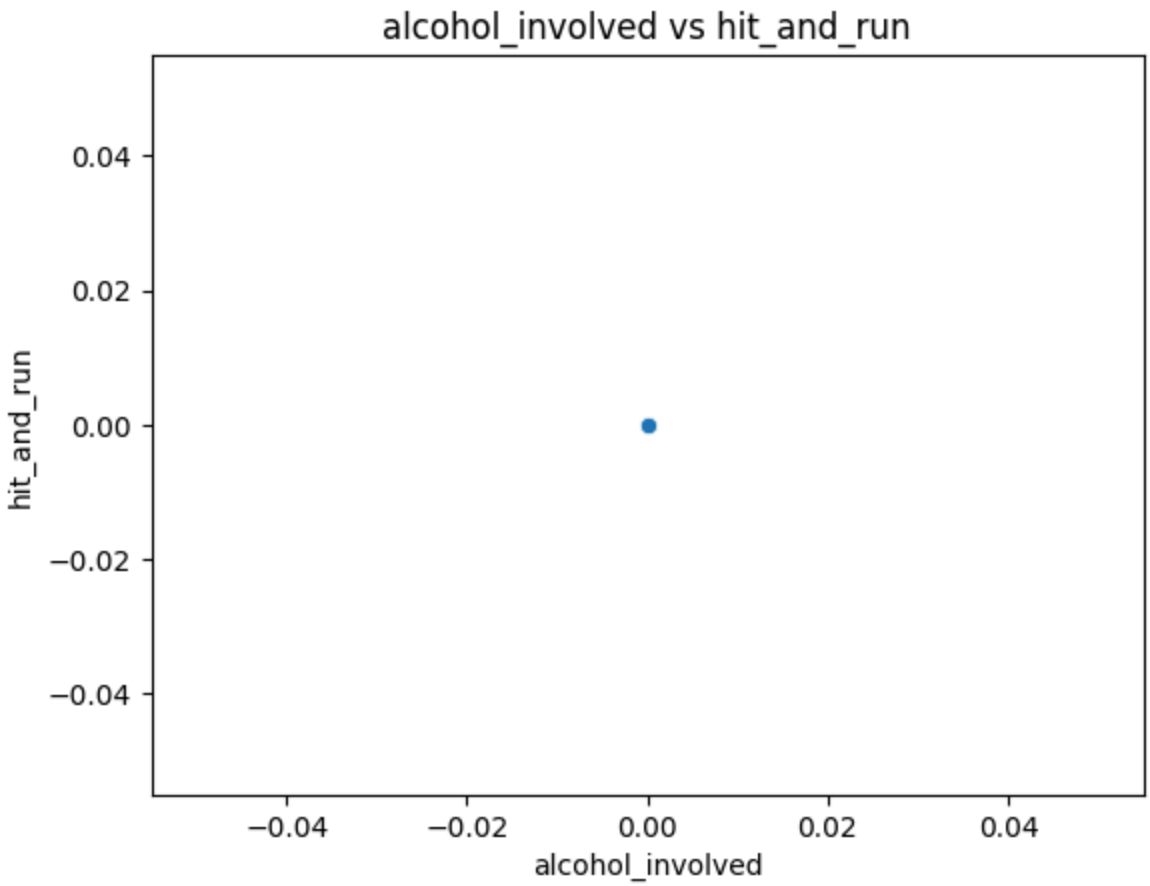

vehicles_involved vs casualties_total
casualties_total
vehicles_involved
3
2
1
0
−1
−0.5
0.0
0.5
1.0
1.5
2.0
2.5

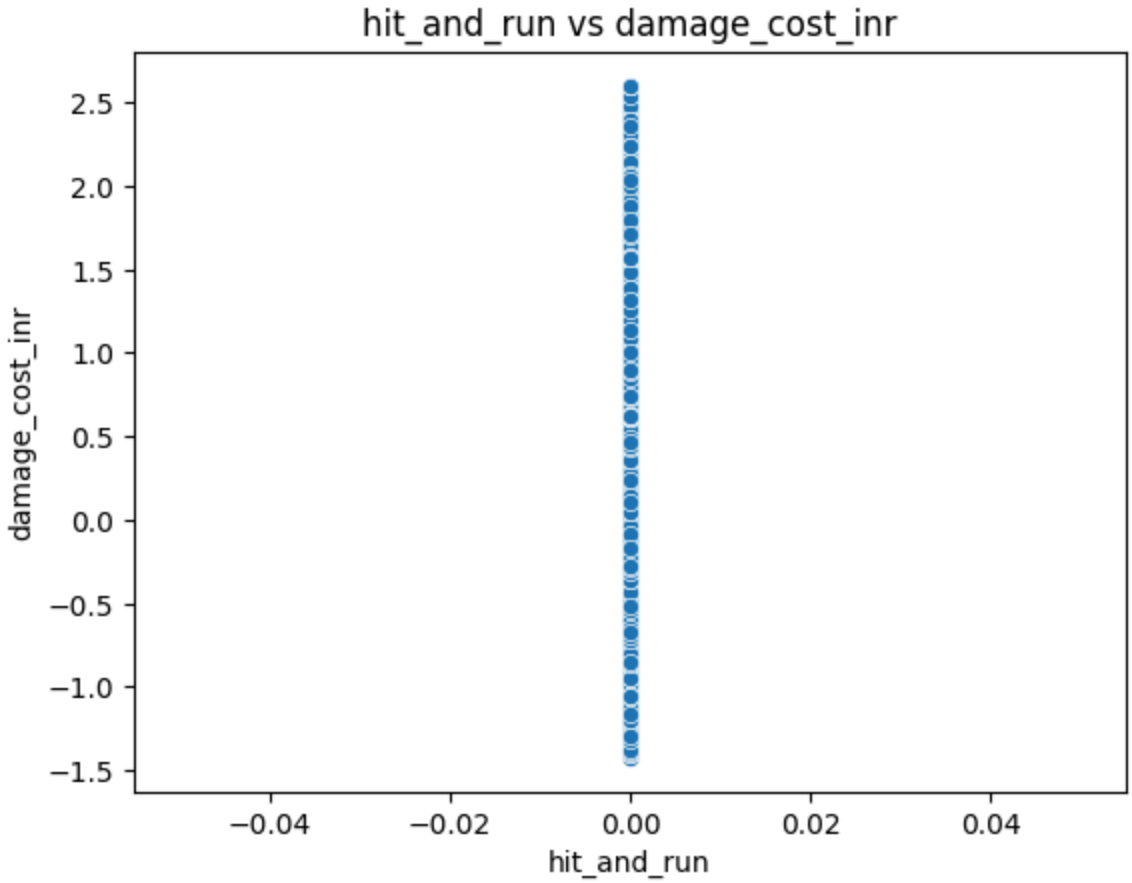

casualties_total vs fatalities
0.04
0.02
0.00
−0.02
−0.04
fatalities
−1
0
1
2
3
casualties_total


fatalities vs serious_injuries
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
serious_injuries
−0.04
−0.02
0.00
0.02
0.04
fatalities


serious_injuries vs minor_injuries
minor_injuries
serious_injuries
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0


minor_injuries vs alcohol_involved
alcohol_involved
minor_injuries
0.04
0.02
0.00
−0.02
−0.04
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


alcohol_involved vs hit_and_run
hit_and_run
alcohol_involved
0.04
0.02
0.00
−0.02
−0.04
−0.04
−0.02
0.00
0.02
0.04


hit_and_run vs damage_cost_inr
damage_cost_inr
hit_and_run
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.04
−0.02
0.00
0.02
0.04

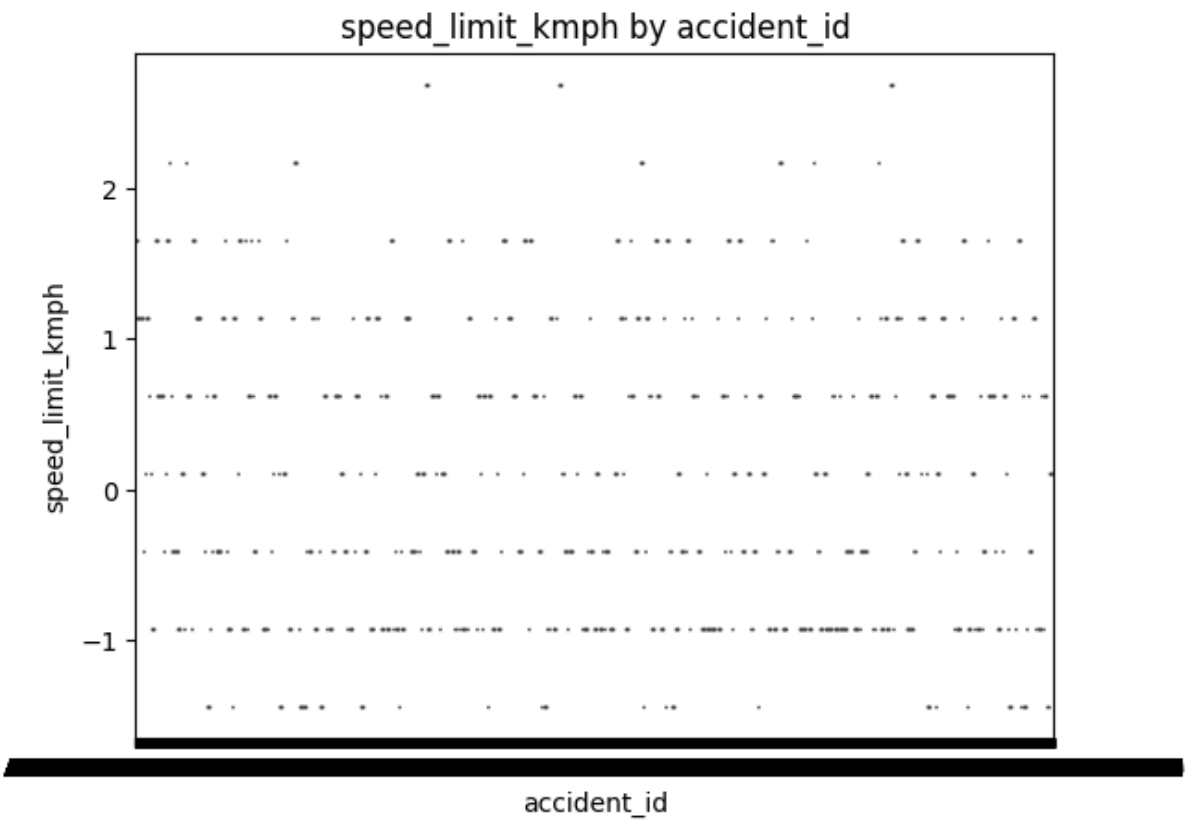
Numerical vs Categorical (Boxplots)

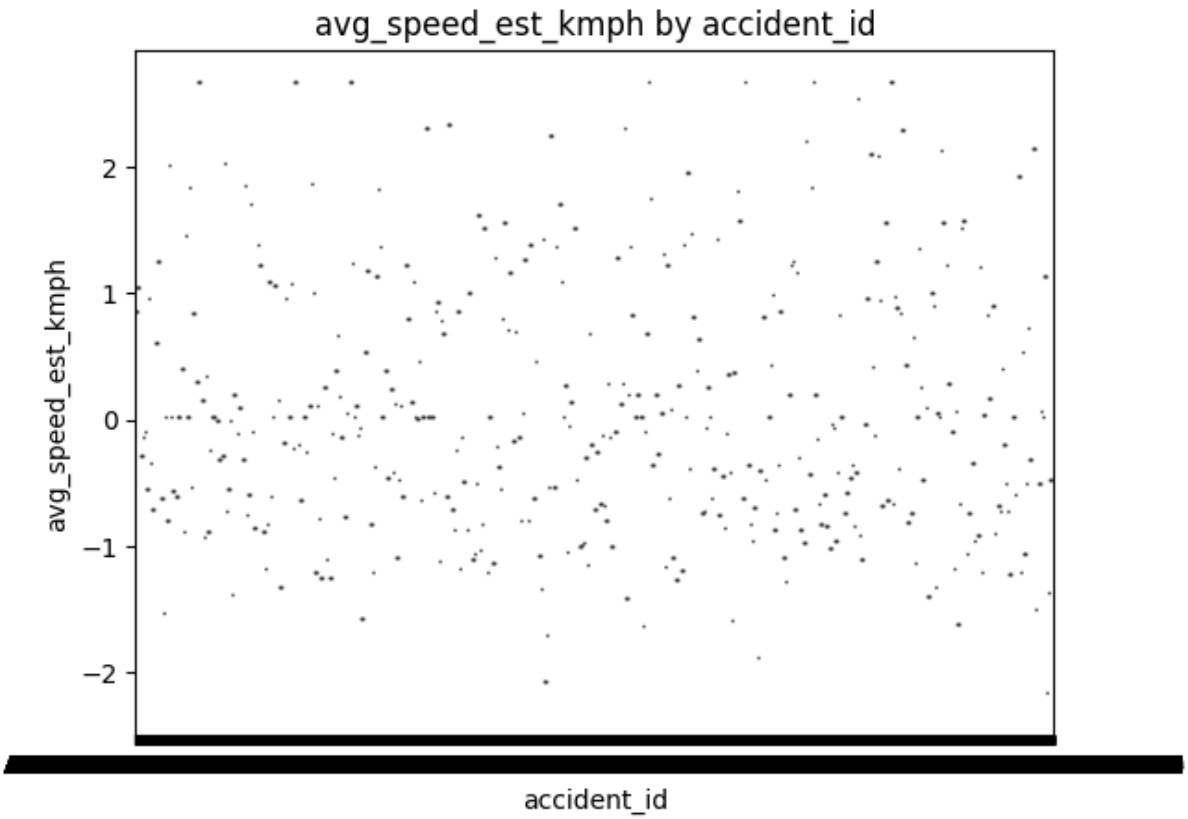
Boxplots comparing numerical variables across categories (e.g., road type) were created.

Insights:

- National highways show higher speed variability.
- Urban roads show lower speed but higher accident density.

In [22]:
```
if len(cat_cols) > 0:
    for col in num_cols[:2]:
        sns.boxplot(
            x=df[cat_cols[0]],
            y=df[col]
        )
        plt.title(f"{col} by {cat_cols[0]}")
        plt.show()
```





---

9. Multivariate Analysis

Pair Plot
A pair plot was created for all numerical variables.

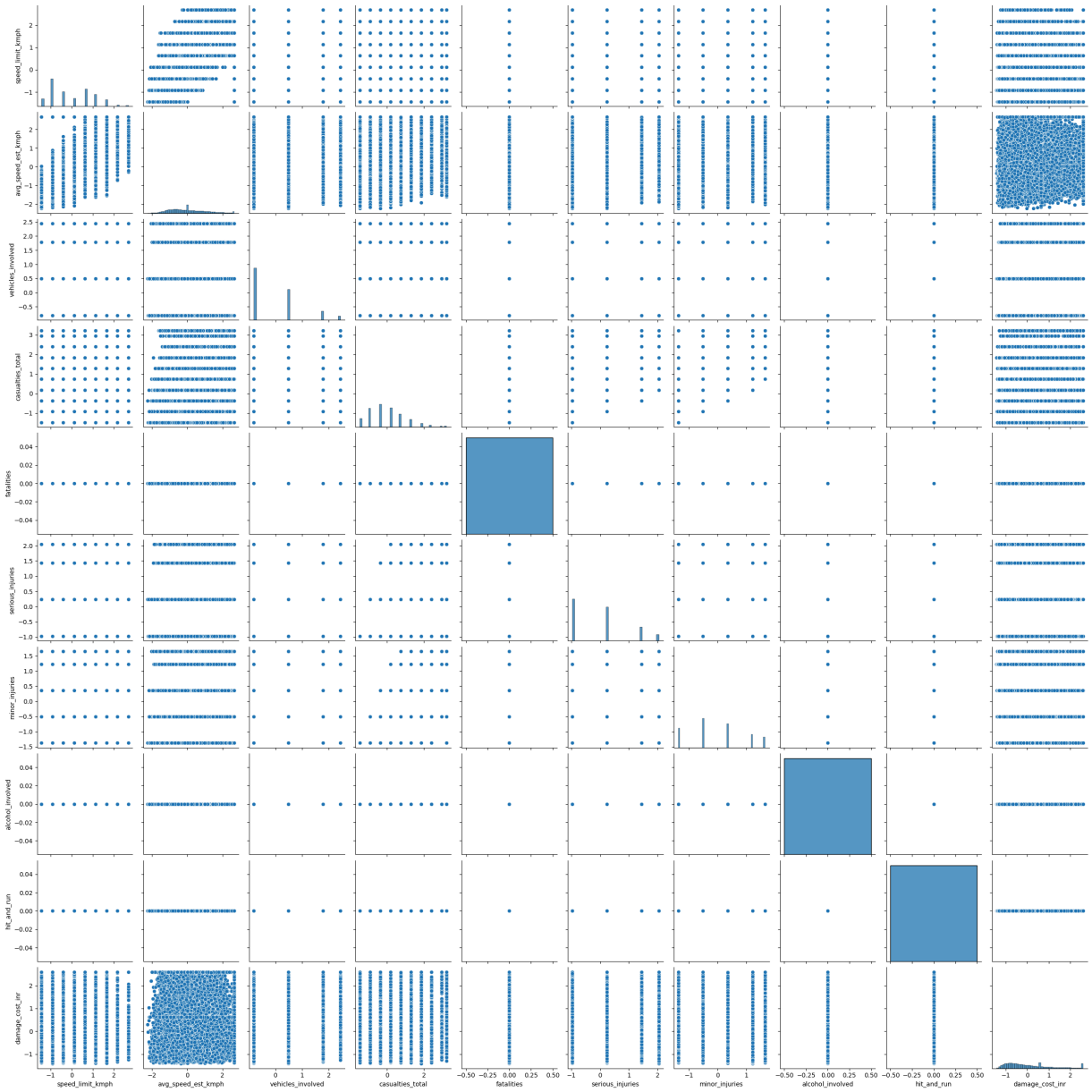
Why pair plot:

- Provides a compact overview of pairwise relationships.
- Helps identify correlated variable pairs visually.

Insight:

- Casualties, injuries, and fatalities form a correlated group.

In [24]:
```
sns.pairplot(df[num_cols])
plt.show()
```



Correlation Heatmap

A heatmap using **sns.heatmap()** was plotted.

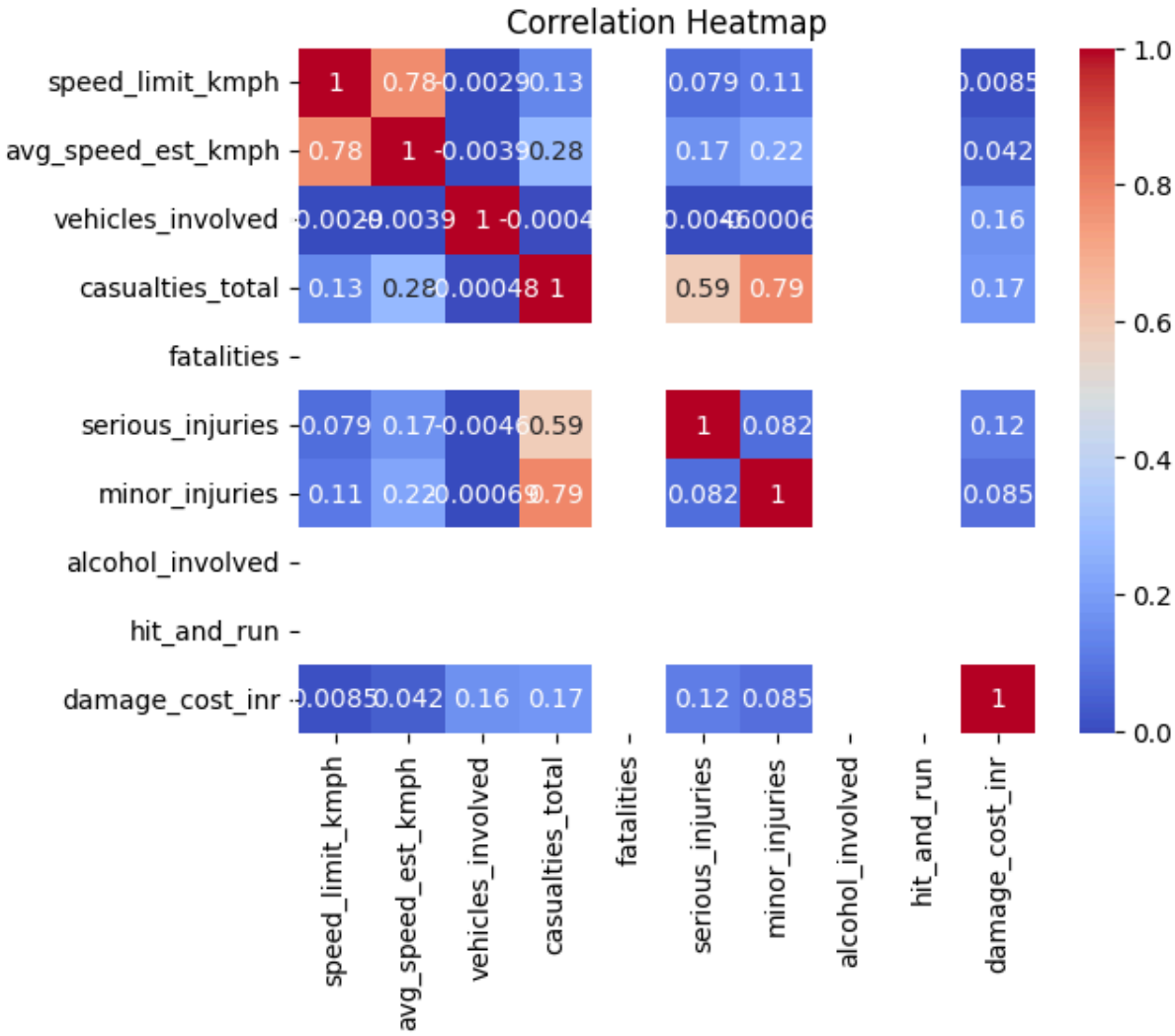
Insights:

- Strong positive correlation between casualties and injuries.
- Speed limit shows moderate correlation with estimated speed.
- Damage cost correlates with injury severity.

Why heatmap:

- Summarizes correlation strength and direction clearly.

In [25]:
```
corr = df[num_cols].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



---

10. Trend Visualization

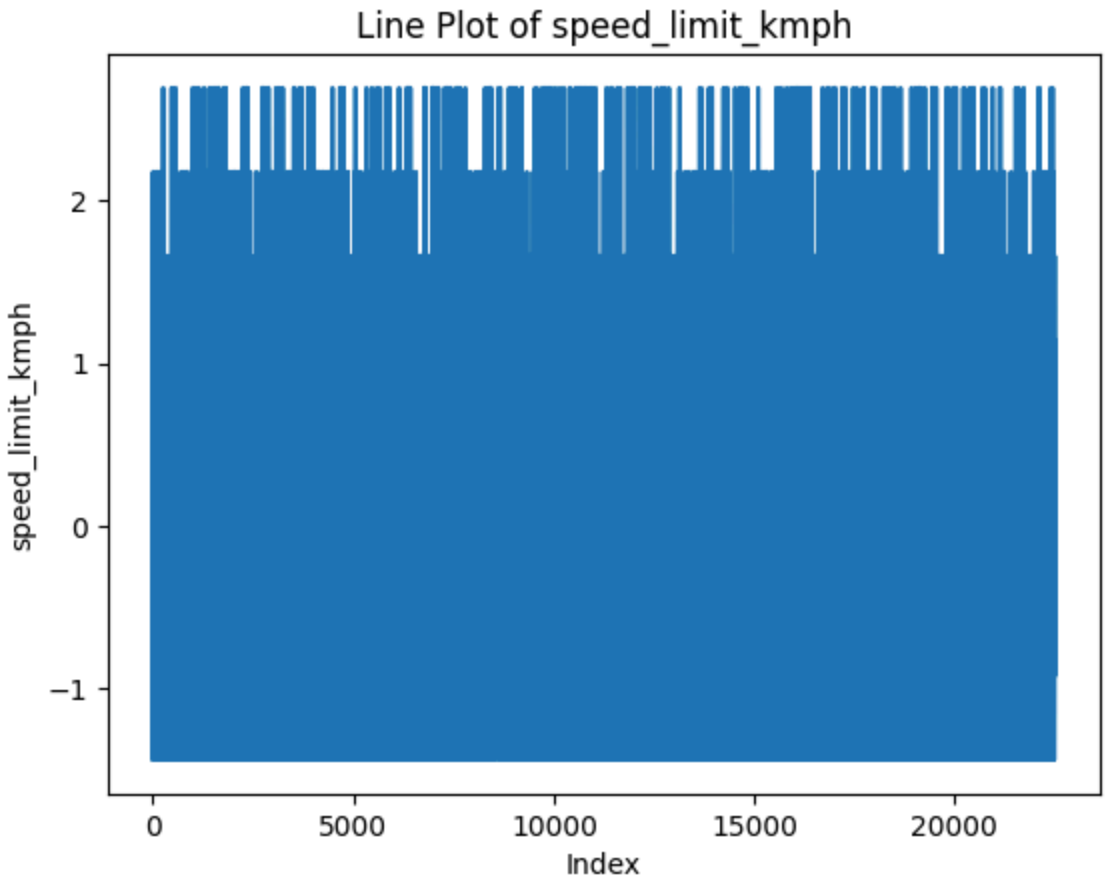
A line plot was generated for a numerical variable across index order.

Purpose:

- To observe overall variability and fluctuation patterns.
- Useful for spotting volatility in accident-related measures.

In [26]:
```
plt.plot(df[num_cols[0]])
plt.title(f"Line Plot of {num_cols[0]}")
plt.xlabel("Index")
```

```
plt.ylabel(num_cols[0])
plt.show()
```



---

11. Probability and Distribution Analysis

Binomial Distribution

A binomial distribution was plotted to demonstrate discrete probability concepts.

Purpose:

- To model event occurrence probability.
- Illustrates probability mass function behavior.

In [27]:
```
from itertools import permutations, combinations

list(permutations([1,2,3], 2))
list(combinations([1,2,3], 2))
```

Out[27]:
```
[(1, 2), (1, 3), (2, 3)]
```

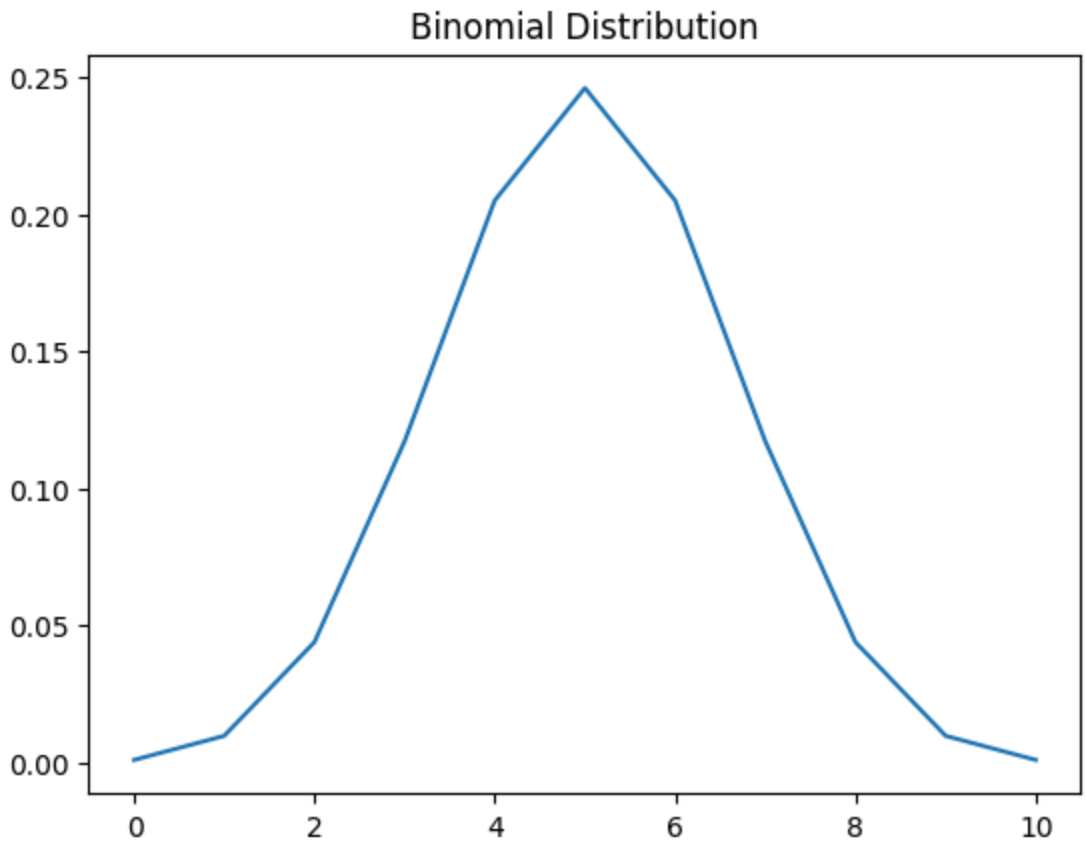
In [28]:
```
P_A = 0.6
P_B = 0.3
```

```
P_A_and_B = P_A * P_B
P_A_or_B = P_A + P_B - P_A_and_B
```

In [29]:
```
values = np.array([0,1])
probs = np.array([0.4,0.6])
np.sum(values * probs)
```

Out[29]:
```
np.float64(0.6)
```

In [30]:
```
x = np.arange(0,11)
plt.plot(x, stats.binom.pmf(x, 10, 0.5))
plt.title("Binomial Distribution")
plt.show()
```



Normal Distribution

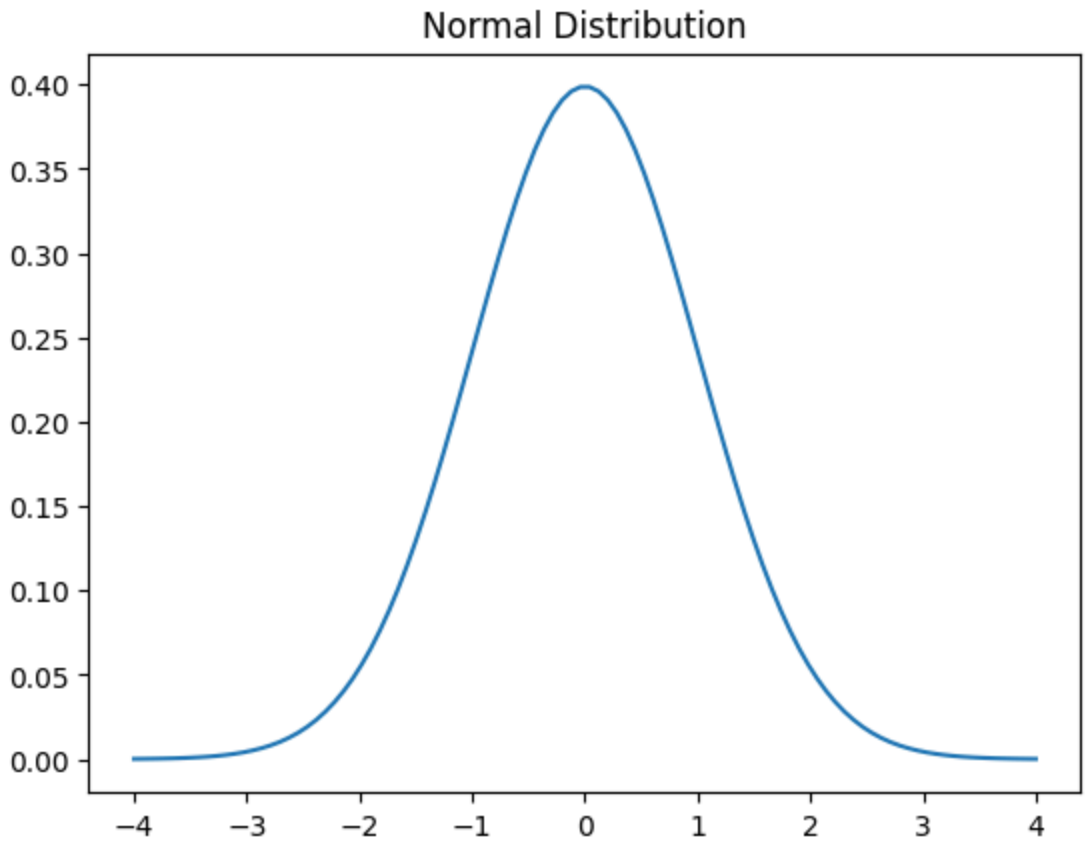
The normal distribution curve was plotted.

Purpose:

- To demonstrate continuous probability distribution.
- Serves as a reference for CLT and hypothesis testing.

In [31]:
```
x = np.linspace(-4,4,100)
plt.plot(x, stats.norm.pdf(x))
```

```
plt.title("Normal Distribution")
plt.show()
```



---

12. Central Limit Theorem Demonstration

Random samples were drawn repeatedly, and their means were plotted.

Insight:

- The sampling distribution of the mean appears approximately normal.
- Confirms the Central Limit Theorem despite original data being skewed.

Importance:

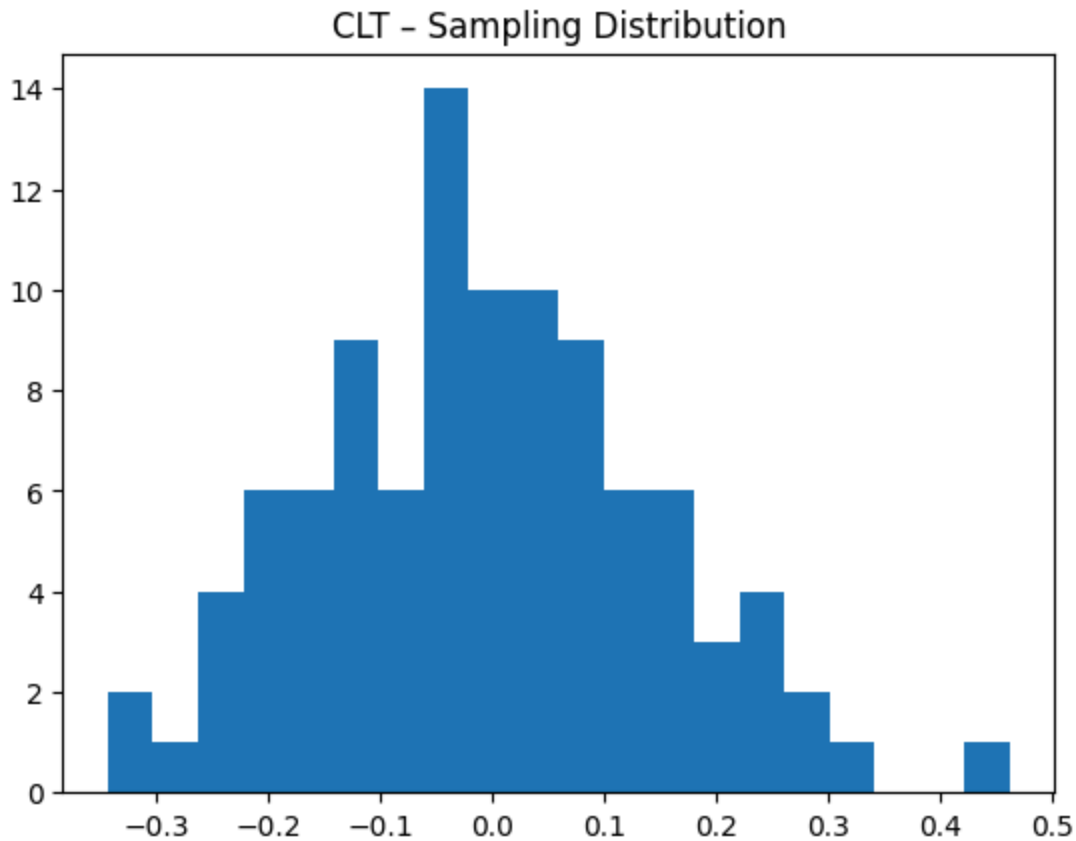
- Justifies use of parametric tests such as t-tests.

In [32]:
```
sample_means = []

for i in range(100):
    sample = df[num_cols[0]].sample(50, replace=True)
    sample_means.append(sample.mean())

plt.hist(sample_means, bins=20)
```

```
plt.title("CLT – Sampling Distribution")
plt.show()
```



---

13. Confidence Interval Estimation

A 95% confidence interval was computed for the mean.

Interpretation:

- We are 95% confident that the true population mean lies within this interval.
- Provides an uncertainty-aware estimate rather than a single value.

In [33]:
```
mean = df[num_cols[0]].mean()
std = df[num_cols[0]].std()
n = len(df)

stats.norm.interval(0.95, mean, std/np.sqrt(n))
```

Out[33]:
```
(np.float64(-0.013066716938314186), np.float64(0.013066716938314415))
```

In [34]:
```
stats.zscore(df[num_cols])
```

Out[34]:
```
array([[-1.43579615, -1.9015961 ,  0.49093896, ...,         nan,
                nan, -0.78117634],
       [-0.92018825, -0.59033319, -0.8074597 , ...,         nan,
                nan,  0.16574998],
       [-0.92018825,  0.02053936,  0.49093896, ...,         nan,
                nan, -0.96356029],
       ...,
       [-0.92018825, -1.13789353, -0.8074597 , ...,         nan,
                nan, -0.66886804],
       [ 0.62663548, -0.81127859, -0.8074597 , ...,         nan,
                nan, -0.50950752],
       [-0.40458034, -0.15804872,  0.49093896, ...,         nan,
                nan,  0.10672592]], shape=(22500, 10))
```

---

14. Hypothesis Testing

One-Sample t-Test

Tested whether the mean differs from a hypothesized value.

Result:

- High p-value → fail to reject null hypothesis.
- Indicates no statistically significant difference from the assumed mean.

In [35]:
```
P_A = 0.3
P_B_given_A = 0.7
P_B = 0.5

(P_B_given_A * P_A) / P_B
```

Out[35]:
```
0.42
```

In [36]:
```
t_stat, p_val = stats.ttest_1samp(df[num_cols[0]], 0)
t_stat, p_val
```

Out[36]:
```
(np.float64(1.7194752087055165e-14), np.float64(0.9999999999999862))
```

In [37]:
```
alpha = 0.05
"Reject H0" if p_val < alpha else "Fail to Reject H0"
```

Out[37]:
```
'Fail to Reject H0'
```

Two-Sample t-Test

Compared two randomly split groups.

Insight:

- No significant difference observed.
- Suggests similar behavior across randomly divided samples.

In [42]:
```
g1, g2 = np.array_split(df[num_cols[0]].sample(frac=1).values, 2)
stats.ttest_ind(g1, g2)
```

Out[42]:
```
TtestResult(statistic=np.float64(-1.213383047434025), pvalue=np.float64(0.22499614
336920504), df=np.float64(22498.0))
```

Chi-Square Test

Tested association between two categorical variables.

Interpretation:

- High p-value indicates no strong dependency.
- Suggests categorical features may be independent.

In [43]:
```
cont_table = pd.crosstab(df[cat_cols[0]], df[cat_cols[1]])
stats.chi2_contingency(cont_table)
```

Out[43]:
```
Chi2ContingencyResult(statistic=np.float64(63135000.00000007), pvalue=np.float64
(0.4013817897089748), dof=63132194, expected_freq=array([[2.66666667e-04, 3.111111
11e-04, 8.88888889e-05, ...,
        4.00000000e-04, 3.55555556e-04, 4.00000000e-04],
       [2.66666667e-04, 3.11111111e-04, 8.88888889e-05, ...,
        4.00000000e-04, 3.55555556e-04, 4.00000000e-04],
       [2.66666667e-04, 3.11111111e-04, 8.88888889e-05, ...,
        4.00000000e-04, 3.55555556e-04, 4.00000000e-04],
       ...,
       [2.66666667e-04, 3.11111111e-04, 8.88888889e-05, ...,
        4.00000000e-04, 3.55555556e-04, 4.00000000e-04],
       [2.66666667e-04, 3.11111111e-04, 8.88888889e-05, ...,
        4.00000000e-04, 3.55555556e-04, 4.00000000e-04],
       [2.66666667e-04, 3.11111111e-04, 8.88888889e-05, ...,
        4.00000000e-04, 3.55555556e-04, 4.00000000e-04]],
      shape=(22500, 2807)))
```

ANOVA
Compared means across groups.

Result:

- No statistically significant difference detected.

In [44]:
```
stats.f_oneway(g1, g2)
```

Out[44]:
```
F_onewayResult(statistic=np.float64(1.472298419800281), pvalue=np.float64(0.224996
14337018475))
```

A/B Testing Simulation
Simulated control vs variant comparison.

Insight:

- Statistically significant difference observed.
- Demonstrates real-world experimentation logic.

In [45]:
```
control = np.random.normal(50,5,100)
variant = np.random.normal(52,5,100)
stats.ttest_ind(control, variant)
```

Out[45]:
```
TtestResult(statistic=np.float64(-2.112401598724526), pvalue=np.float64(0.03590537
181891341), df=np.float64(198.0))
```

---

15. Final Conclusion

This project successfully applied exploratory data analysis, visualization techniques, and statistical inference to a large-scale road accident dataset. The analysis revealed key insights into accident severity, speed behavior, and injury patterns. Visualizations enabled intuitive understanding, while inferential statistics provided mathematical validation.

The results highlight that while most accidents are minor, extreme cases significantly impact fatalities and economic damage. Such findings can support targeted road safety interventions, policy planning, and further predictive modeling.